# Executive Summary of Cook Inlet Beluga eDNA Results

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**Project Methods Overview**

These are the results of a pilot study of eDNA in Cook Inlet from Nov. 2020 to Nov. 2021. This pilot project consisted of two eDNA sample collections paired with visual CIB surveys:

1. Monthly samples were collected at two sites at two depths each (0m & 5m) in the Kenai River across the year,
2. Higher frequency sampling (sub-weekly/biweekly) of 20 Mile River was conducted from August 2021-Nov. 2021 at the surface (0m)

We successfully designed and implemented sample collection and processing methodologies to track Beluga and their prey. We specifically applied qPCR assays for Beluga, Coho, Chinook, and Eulachon (*Thaleichthys pacificus*) and vertebrate community metabarcoding assays (MiFish 12S) to all eDNA samples. All samples were collected at high tide when Beluga are known to enter river systems (Moore et al. 2000).

## Figure 1. Map of Cook Inlet Sampling Locations

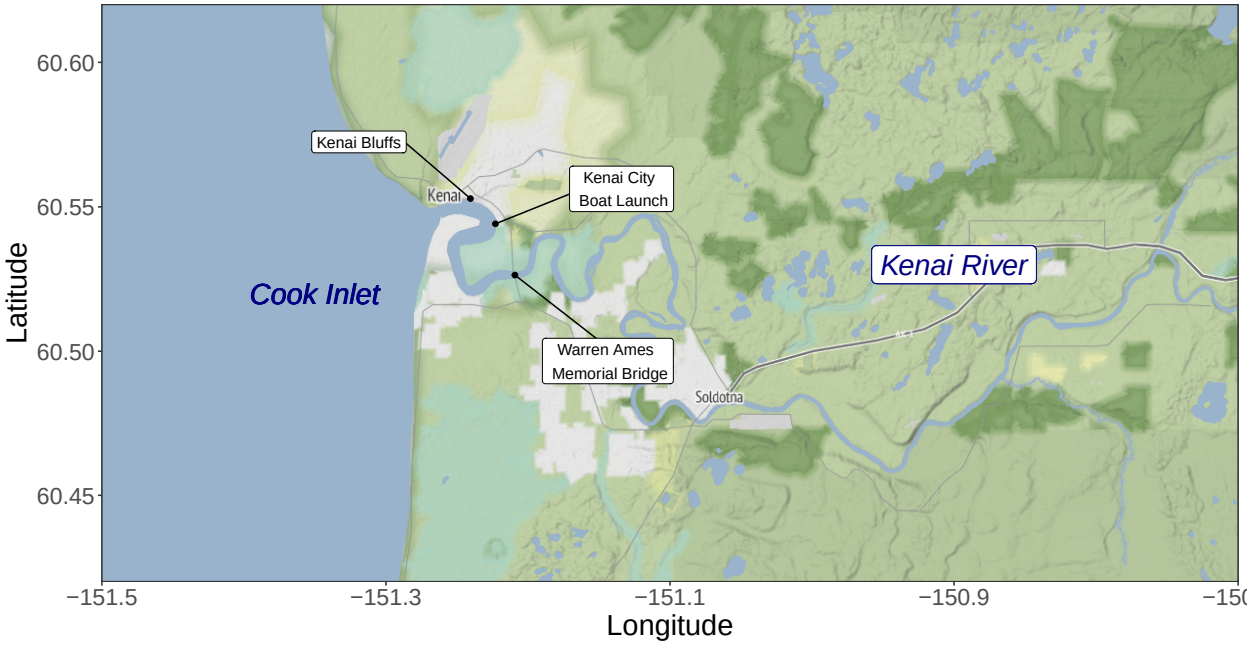
**Cook Inlet Inset map**

```
## i © Stadia Maps © Stamen Design © OpenMapTiles © OpenStreetMap contributors.
```

```
## i 80 tiles needed, this may take a while (try a smaller zoom?)
```

**Kenai Inset map**

```
## i © Stadia Maps © Stamen Design © OpenMapTiles © OpenStreetMap contributors.
```

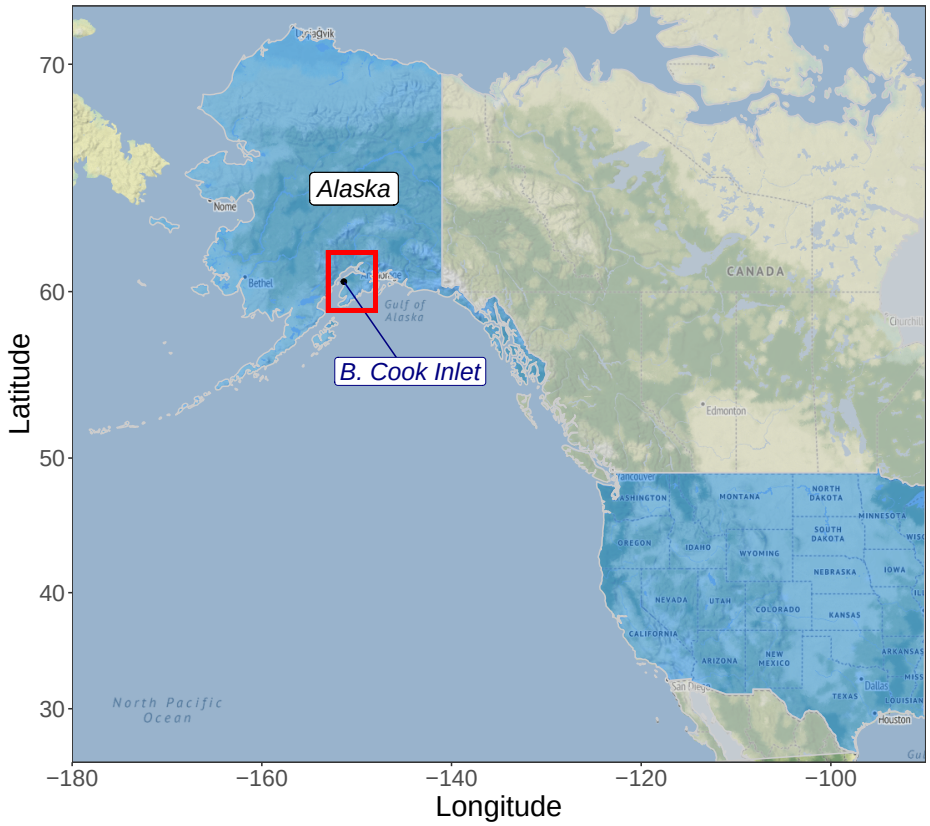



### Turnagain Inset map

```
## i © Stadia Maps © Stamen Design © OpenMapTiles © OpenStreetMap contributors.
```

### Alaska wide map

```
## i © Stadia Maps © Stamen Design © OpenMapTiles © OpenStreetMap contributors.
```



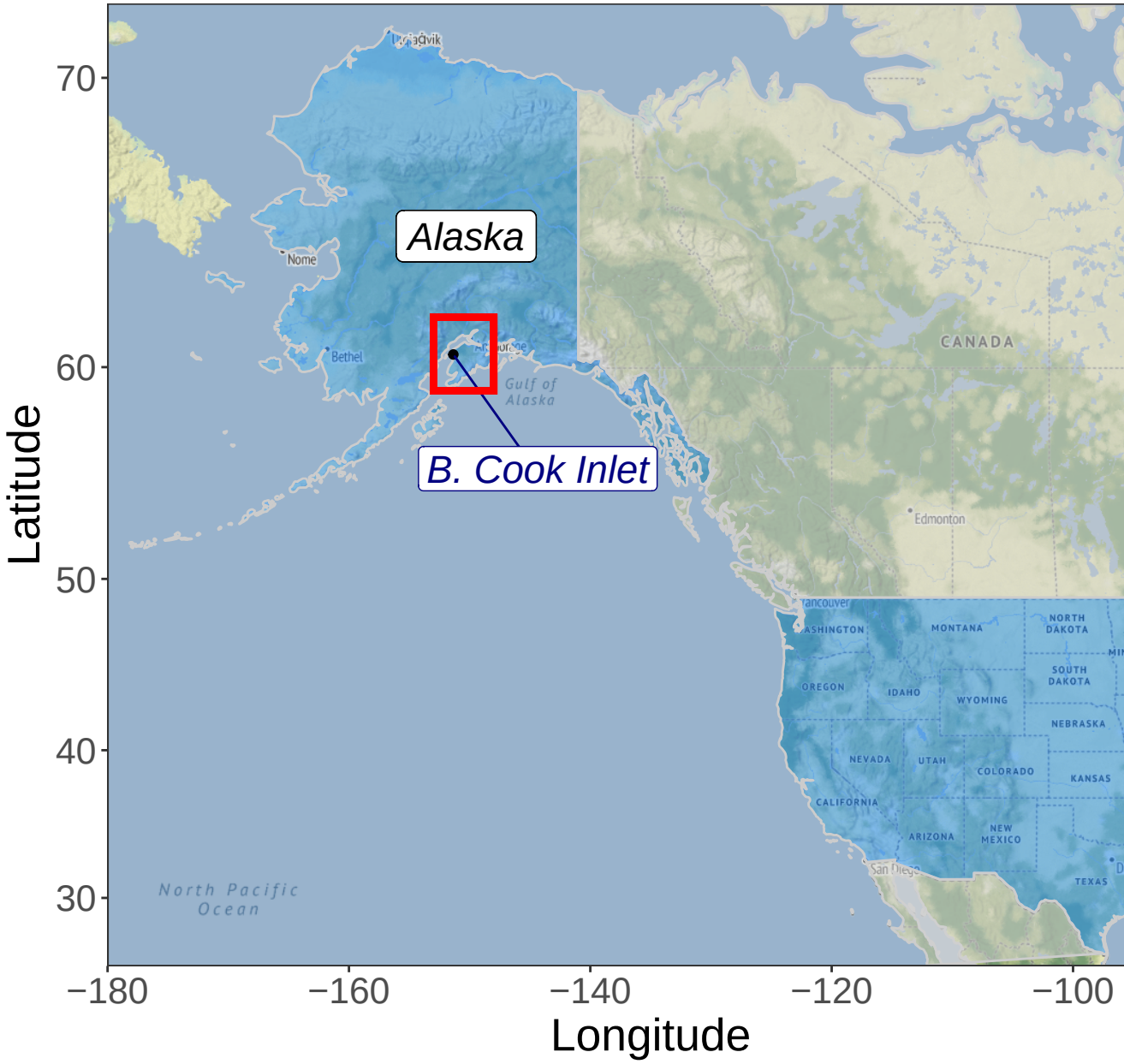

```
layout <- c(
  patchwork::area(1, 1, 8,8),
  patchwork::area(1, 9, 8, 16),
  patchwork::area(9, 1, 12,8),
  patchwork::area(9, 9, 12, 16))

# Show the layout to make sure it looks as it should
#plot(layout)
```

**Layout Map**

**Plot Combined Figure 1**

A


Alaska
B. Cook Inlet
Nome
Bethel
Gulf of Alaska
North Pacific Ocean
CANADA
Edmonton
Vancouver
WASHINGTON
MONTANA
NORTH DAKOTA
SOUTH DAKOTA
OREGON
IDAHO
WYOMING
NEBRASKA
NEVADA
UTAH
COLORADO
KANSAS
CALIFORNIA
ARIZONA
NEW MEXICO
TEXAS
San Diego
70
60
50
40
30
Latitude
−180
−160
−140
−120
−100
Longitude

C


Turnagain Arm
93.5 Mile Marker
Bird Point
61.00
60.95
60.90
atitude

# Summary of Major Findings

**Twentymile River time series provides valuable input on identifying critical foraging habitat for CIB.**

- Weekly time series before and during beluga foraging events provides sufficient resolution to correlate changes in predator occurrence and prey field shifts.
- We demonstrate the importance of implementing a quantitative eDNA framework based on both metabarcoding + qPCR data.

In Twentymile River, we had congruent results of absolute DNA abundance changes (qPCR) and relative abundance changes (metabarcoding - both bias corrected and transformed) with increased abundance of Coho salmon occurring during beluga occurrences in Twentymile river. Importantly, in Twentymile River we happened to implement a qPCR assay specifically targeting the multiple species that alternated in abundance over the time series, allowing us to quantitatively estimate fish abundances.

- **eDNA sampling paired with visual surveys across major known Beluga estuary/river aggregation sites can directly inform Stochastic Dynamic Programming models of Cook Inlet Belugas which are critical for accurately evaluating Population Consequences of Disturbance on this Endangered species.**

**Informing Accurate eDNA Based Prey Field Identification of Cook Inlet Belugas**

1. Visual observations have a higher detection rate of Beluga whales than eDNA approaches, more consistently detecting Belugas during periods of river utilization. However, our results demonstrate that **both qPCR and metabarcoding eDNA approaches can successfully detect Beluga whales, albeit less consistently than visual results**. These results are particularly important for extending monitoring efforts beyond the current AKBMP monitoring window and sites with particular value in Winter months and remote sites that are difficult to frequently access, opening the door to autonomous eDNA detection of Cook Inlet Beluga whales.

2. In high tidal flow riverine systems, we find minimal added benefit in sampling multiple sites and depths along the river. Thus we strongly **recommend a reallocation of sampling effort, focusing on a single site sampled more frequently**.

3. Our results demonstrate the power of quantitative metabarcoding frameworks for the accurate interpretation of observed eDNA signatures. The framework employed in this study leveraged three data streams 1) qPCR assays which provide absolute abundance estimates, 2) metabarcoding data which provide biased proportional abundances, and 3) mock communities to correct for the amplification biases impacting metabarcoding data. Successful application of these approaches requires:

    1) qPCR assay(s) targeting the most abundant species observed in eDNA samples,
    2) Reliable mock community data for target species, and
    3) No swamping of metabarcoding signals.

    Thus, we recommend the application of **additional Sockeye, Pink, and Chum salmon qPCR measurements in addition to Eulachon, Coho, King, and Beluga qPCR assays** as well as a **more comprehensive mock community dataset** to provide valuable insights to inform future quantitative metabarcoding applications identified above.

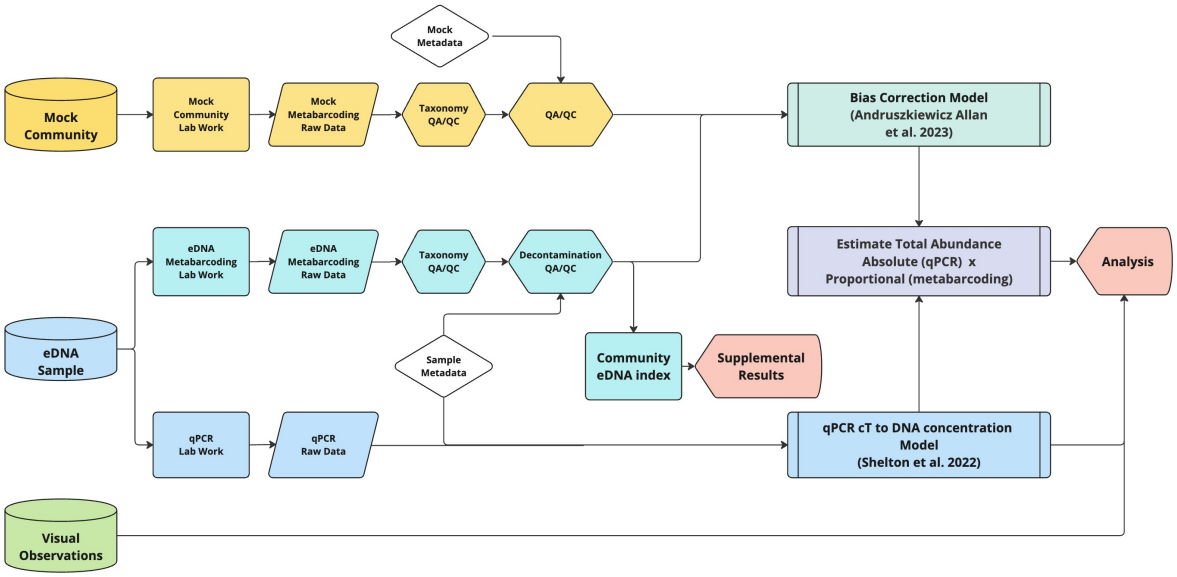



Figure 1: A caption

# Figure 2. Workflow of Quantitative Metabarcoding

4. Although we demonstrate successful eDNA sample processing, the high sediment load of Cook Inlet waters posed a significant challenge to water filtration. Here we deployed 1 µm filters to filter 1 L samples which took up to 2 hours to filter given the high sediment load (0.6µm sediment). For future deployments **we recommend the utilization of 5µm eDNA filters** which should ease the filtration burden and have been found to collect sufficient marine vertebrate eDNA (Power et al. 2023).

# Visual, Metabarcoding, & qPCR Detection Comparisons

## Twentymile River Results

**Note this is the focus of the study and the highlight results.**

**Beluga Observations: Visual vs. eDNA Metabarcoding vs. qPCR.** eDNA and visual observations of Cook Inlet Beluga whales were largely concordant within Turnagain Arm (Figure 1). eDNA detections at 20 Mile River occurred during when Beluga were only visually observed at 93.5 Mile Marker and were heard to be along the 1 West of Hope (~20km from the mouth of Twentymile River). This strongly suggests that Beluga eDNA is being transported >20 km by tidal bores in the second largest tidal exchange system on Earth.

**Figure 3. Visual and eDNA detections of beluga occurrence in Turnagain Arm.**

### General Statistics for qPCR data

```
## [1] 258
```

```
## # A tibble: 4 x 5
##   `Target Name`   NAs detections non_detections total_samples
##   <chr>         <int>      <int>          <int>         <int>
## 1 Chinook           0        106            152           258
## 2 Coho              0        213             45           258
## 3 Eulachon          0        204             54           258
## 4 cetacea           0         56            199           255
```

```
## # A tibble: 4 x 5
##   `Target Name` `mean(mean_mean_dnacopy_L)` `max(mean_mean_dnacopy_L)`
##   <chr>                               <dbl>                      <dbl>
## 1 Chinook                             2068.                     15530.
## 2 Coho                               31070.                    338892.
## 3 Eulachon                           17464.                    627828.
## 4 cetacea                             3140.                     51775.
## # i 2 more variables: `sd(mean_mean_dnacopy_L)` <dbl>,
## #   `n_distinct(mean_mean_dnacopy_L)` <int>
```

**Pull in metabarcoding data**

**General Statistics for Metabarcoding data**

```
## [1] 229
```

```
## [1] 163
```

```
## # A tibble: 10,020 x 1
##    total_sum
##        <dbl>
##  1  15340132
##  2  15340132
##  3  15340132
##  4  15340132
##  5  15340132
##  6  15340132
##  7  15340132
##  8  15340132
##  9  15340132
## 10  15340132
## # i 10,010 more rows
```

```
## [1] 64
```

```
## [1] 18
```

```
## [1] 31
```

```
## [1] 12
```

```
## [1] ""                 "Beringraja rhina" "Squalus suckleyi"
```

```
## [1] ""                          "Entosphenus tridentatus"
```

```
## [1] "Actinopteri"    "Mammalia"       "Aves"           "Chondrichthyes"
## [5] "Salmoniformes"  "Hyperoartia"
```

```
## # A tibble: 110 x 36
## # Groups:   New_name [2]
##    sum.taxonomy     Domain Phylum Class Order Family Genus Species Sample nReads
##    <chr>            <chr>  <chr>  <chr> <chr> <chr>  <chr> <chr>   <chr>   <dbl>
##  1 Eukaryota;Chord~ Eukar~ Chord~ Acti~ Gadi~ Gadid~ Gadus ""      c19_f~      0
##  2 Eukaryota;Chord~ Eukar~ Chord~ Acti~ Gadi~ Gadid~ Gadus ""      c19_f~      0
##  3 Eukaryota;Chord~ Eukar~ Chord~ Aves  Acci~ Accip~ Hali~ "Halia~ c19_f~      0
##  4 Eukaryota;Chord~ Eukar~ Chord~ Aves  Acci~ Accip~ Hali~ "Halia~ c19_f~      0
##  5 Eukaryota;Chord~ Eukar~ Chord~ Aves  Pass~ Corvi~ Corv~ "Corvu~ c19_f~      0
##  6 Eukaryota;Chord~ Eukar~ Chord~ Aves  Pass~ Corvi~ Corv~ "Corvu~ c19_f~      0
##  7 Eukaryota;Chord~ Eukar~ Chord~ Acti~ Pleu~ Pleur~ Plat~ "Plati~ c19_f~   3753
##  8 Eukaryota;Chord~ Eukar~ Chord~ Acti~ Pleu~ Pleur~ Plat~ "Plati~ c19_f~      0
##  9 Eukaryota;Chord~ Eukar~ Chord~ Mamm~ Euli~ Soric~ Sorex "Sorex~ c19_f~      0
## 10 Eukaryota;Chord~ Eukar~ Chord~ Mamm~ Euli~ Soric~ Sorex "Sorex~ c19_f~      0
## # i 100 more rows
## # i 26 more variables: New_name <chr>, Seq_number <chr>,
## #   New_name_no_tech <chr>, LabID <chr>, Station <chr>, Site <chr>,
## #   Bio_rep <int>, Tech_rep <int>, Sample_date <date>, Sample_Control <chr>,
## #   Control_Type <chr>, High_Tide_.Time <chr>, Tide_height <dbl>,
## #   Time_of_Collection <chr>, Beluga_Observed <chr>, Beluga <chr>,
## #   Date_Sampled <chr>, River <chr>, Depth <chr>, Volume <dbl>, ...
```

```
## [1] 19283
```

```
## [1] 220382
```

```
## [1] 65764.53
```

```
## # A tibble: 2 x 1
##   New_name
##   <chr>
## 1 Dock_5m_Nov.2020_3
## 2 Dock_Surface_Jan_3
```

```
combined_data_long_reads_c %>%
  mutate(., salmon = if_else(str_detect(sum.taxonomy,"Oncorhynchus"),"salmon","other")) %>%
  filter(., River == "Twentymile") %>%
  mutate(., sum.total_reads = sum(nReads)) %>%
  group_by(salmon,sum.total_reads) %>%
  dplyr::summarise(group_reads = sum(nReads)) %>%
  mutate(., perc=group_reads/sum.total_reads)
```

```
## `summarise()` has regrouped the output.
## i Summaries were computed grouped by salmon and sum.total_reads.
## i Output is grouped by salmon.
## i Use `summarise(.groups = "drop_last")` to silence this message.
## i Use `summarise(.by = c(salmon, sum.total_reads))` for per-operation grouping
##   (`?dplyr::dplyr_by`) instead.
```

```
## # A tibble: 2 x 4
## # Groups:   salmon [2]
##   salmon sum.total_reads group_reads  perc
##   <chr>            <dbl>       <dbl> <dbl>
## 1 other          9208065     1554134 0.169
## 2 salmon         9208065     7653931 0.831
```

```
# Filter for only species observed with qPCR
combined_data_long_reads_c %>%
  filter(., Species %in% c("Delphinapterus leucas", "Thaleichthys pacificus","Oncorhynchus tshawytscha"

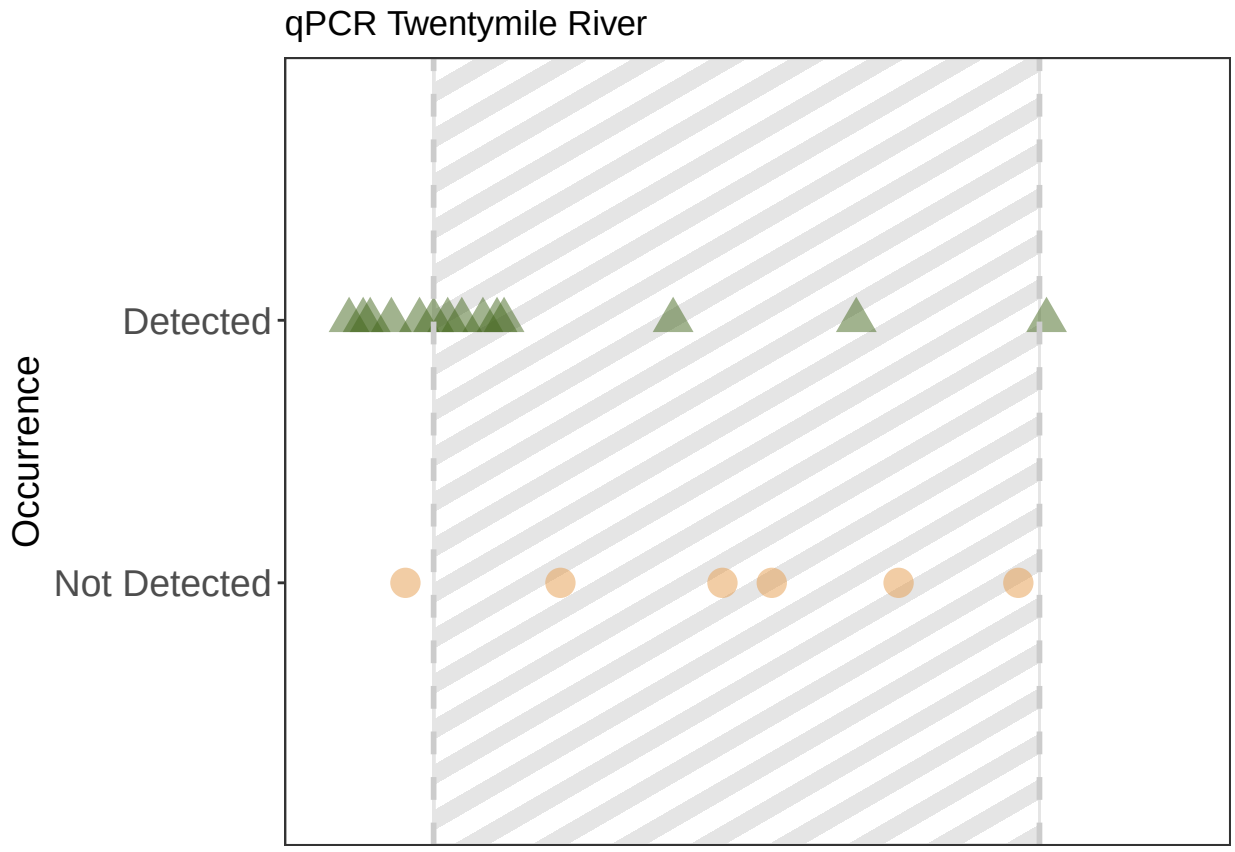
meta_data_only_qPCR_species_long %>%
  mutate(., Sample_date = as.Date(Sample_date,"%m/%d/%y")) %>%
  mutate(., Tide_height = as.numeric(Tide_height)) %>%
  left_join(combined_data_long_reads_c) %>%
    mutate(., `Target Name`=case_when(Species=="Delphinapterus leucas"~ "cetacea" ,
                                      Species=="Thaleichthys pacificus" ~ "Eulachon",
                                      Species=="Oncorhynchus tshawytscha"~ "Chinook",
                                      Species=="Oncorhynchus kisutch" ~ "Coho")) %>%
  left_join(qPCR_4_species) %>%
  mutate(., `Target Name`=if_else(`Target Name`=="cetacea","Beluga",`Target Name`))-> combo_4_species
```

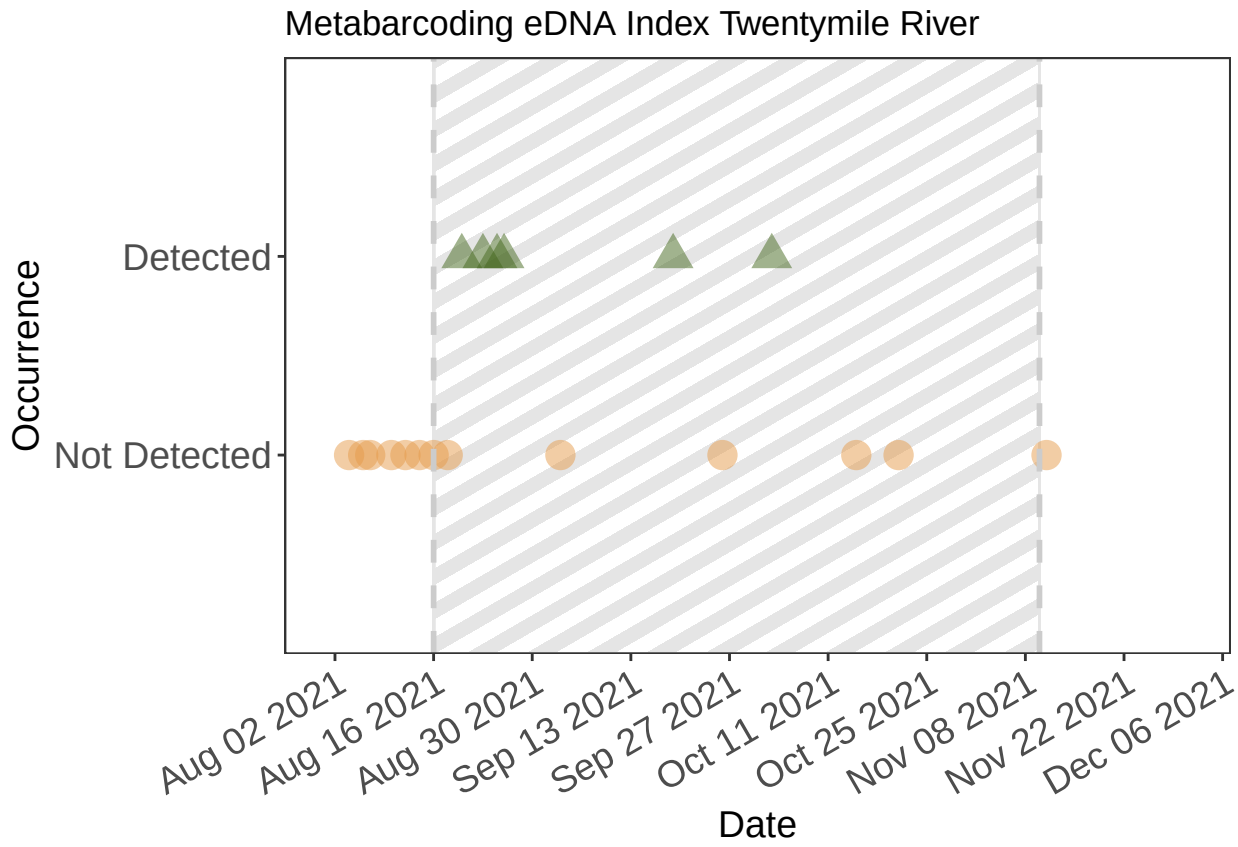
```
## Joining with ‘by = join_by(sum.taxonomy, Domain, Phylum, Class, Order, Family,
## Genus, Species, Sample, nReads, New_name, Seq_number, New_name_no_tech, LabID,
## Station, Site, Bio_rep, Tech_rep, Sample_date, Sample_Control, Control_Type,
## High_Tide_.Time, Tide_height, Time_of_Collection, Beluga_Observed, Beluga,
## Date_Sampled, River, Depth, Volume, ID_mifish, total_reads, prop_reads,
## max_prop, index_scores)‘
## Joining with ‘by = join_by(New_name_no_tech, LabID, Site, Bio_rep, Sample_date,
## Sample_Control, Control_Type, High_Tide_.Time, Tide_height, Time_of_Collection,
## Beluga_Observed, Beluga, Date_Sampled, River, Depth, Volume, ‘Target Name‘)‘

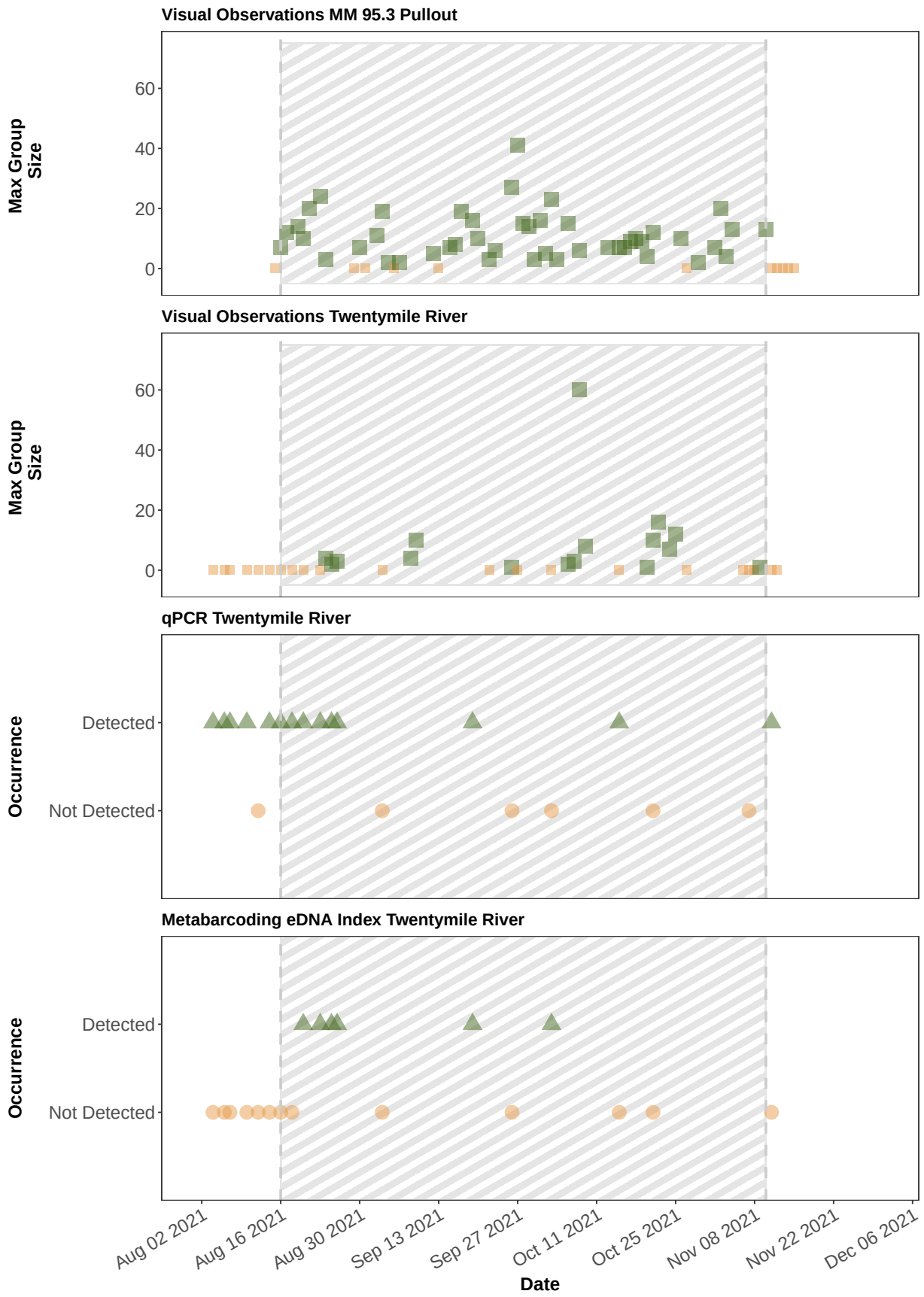
## Warning in left_join(., qPCR_4_species): Detected an unexpected many-to-many relationship between ‘x
## i Row 1 of ‘x‘ matches multiple rows in ‘y‘.
## i Row 183 of ‘y‘ matches multiple rows in ‘x‘.
## i If a many-to-many relationship is expected, set ‘relationship =
##   "many-to-many"‘ to silence this warning.
```

```
saveRDS(combo_4_species, here("analysis","combo_4_species.RDS"))
```

### Organizing actual plots


qPCR Twentymile River
Occurrence
Detected
Not Detected


Metabarcoding eDNA Index Twentymile River
Occurrence
Detected
Not Detected
Aug 02 2021
Aug 16 2021
Aug 30 2021
Sep 13 2021
Sep 27 2021
Oct 11 2021
Oct 25 2021
Nov 08 2021
Nov 22 2021
Dec 06 2021
Date


Visual Observations MM 95.3 Pullout
Max Group Size
60
40
20
0
Visual Observations Twentymile River
Max Group Size
60
40
20
0
qPCR Twentymile River
Occurrence
Detected
Not Detected
Metabarcoding eDNA Index Twentymile River
Occurrence
Detected
Not Detected
Aug 02 2021
Aug 16 2021
Aug 30 2021
Sep 13 2021
Sep 27 2021
Oct 11 2021
Oct 25 2021
Nov 08 2021
Nov 22 2021
Dec 06 2021
Date

## Figure 4.

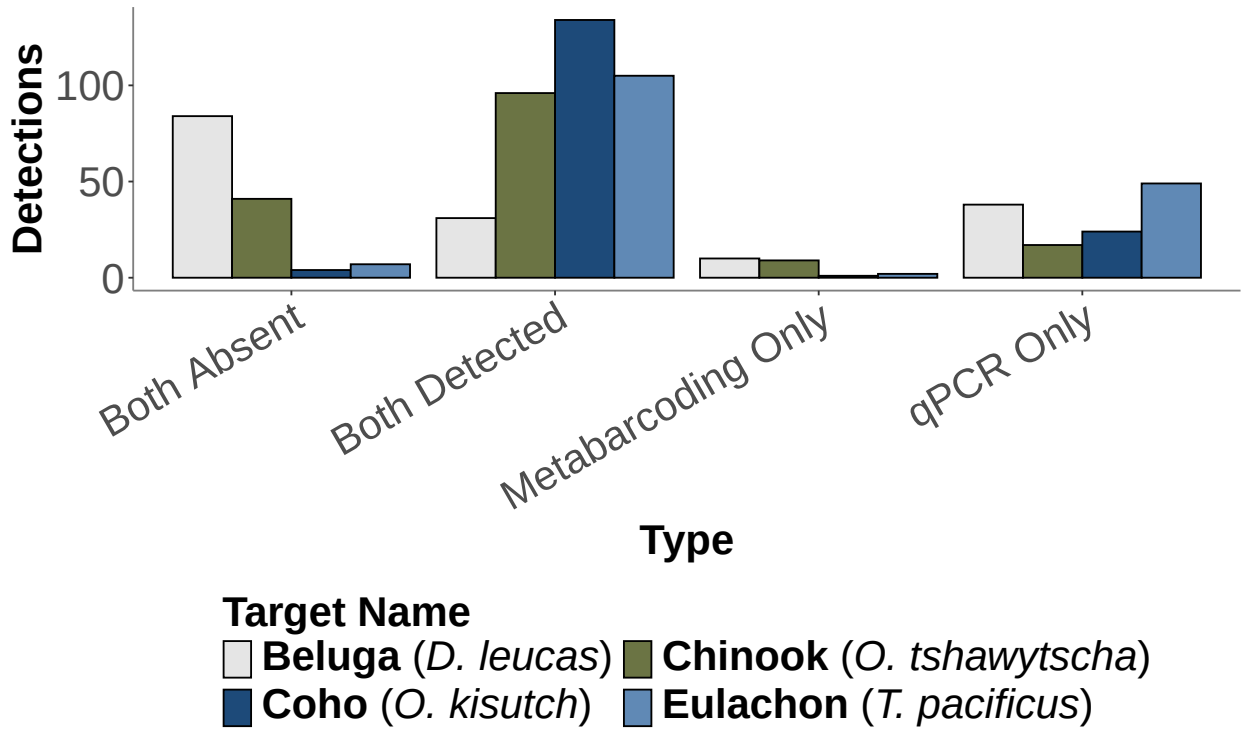
```
## # A tibble: 8 x 3
## # Groups:   Target Name [4]
##   `Target Name` qPCR_PA     n
##   <chr>           <dbl> <int>
## 1 Beluga              0   159
## 2 Beluga              1    98
## 3 Chinook             0   128
## 4 Chinook             1   130
## 5 Coho                0    32
## 6 Coho                1   226
## 7 Eulachon            0    43
## 8 Eulachon            1   215
```

```
## # A tibble: 8 x 3
## # Groups:   Target Name [4]
##   `Target Name` metabarcoding_PA     n
##   <chr>                    <dbl> <int>
## 1 Beluga                       0   122
## 2 Beluga                       1    41
## 3 Chinook                      0    58
## 4 Chinook                      1   105
## 5 Coho                         0    28
## 6 Coho                         1   135
## 7 Eulachon                     0    56
## 8 Eulachon                     1   107
```

```
## # A tibble: 16 x 4
## # Groups:   Target Name [4]
##    `Target Name` Detection              n tot_obs
##    <chr>         <chr>              <int>   <int>
##  1 Beluga        Both Absent           84     163
##  2 Beluga        Both Detected         31     163
##  3 Beluga        Metabarcoding Only    10     163
##  4 Beluga        qPCR Only             38     163
##  5 Chinook       Both Absent           41     163
##  6 Chinook       Both Detected         96     163
##  7 Chinook       Metabarcoding Only     9     163
##  8 Chinook       qPCR Only             17     163
##  9 Coho          Both Absent            4     163
## 10 Coho          Both Detected        134     163
## 11 Coho          Metabarcoding Only     1     163
## 12 Coho          qPCR Only             24     163
## 13 Eulachon      Both Absent            7     163
## 14 Eulachon      Both Detected        105     163
## 15 Eulachon      Metabarcoding Only     2     163
## 16 Eulachon      qPCR Only             49     163
```

```
## # A tibble: 12 x 5
## # Groups:   Target Name [4]
##    `Target Name` Detection              n tot_obs    perc
##    <chr>         <chr>              <int>   <int>   <dbl>
```

```
##  1 Beluga        Both                115     163 0.706
##  2 Beluga        Metabarcoding Only   10     163 0.0613
##  3 Beluga        qPCR Only            38     163 0.233
##  4 Chinook       Both                137     163 0.840
##  5 Chinook       Metabarcoding Only    9     163 0.0552
##  6 Chinook       qPCR Only            17     163 0.104
##  7 Coho          Both                138     163 0.847
##  8 Coho          Metabarcoding Only    1     163 0.00613
##  9 Coho          qPCR Only            24     163 0.147
## 10 Eulachon      Both                112     163 0.687
## 11 Eulachon      Metabarcoding Only    2     163 0.0123
## 12 Eulachon      qPCR Only            49     163 0.301
```




```
metabar_detections_bottle %>%
  filter(., str_detect(New_name_no_tech, "TM")) -> metabar_detections_bottle_TM

qPCR_detections_bottle %>%
  filter(., str_detect(New_name_no_tech, "TM")) -> qPCR_detections_bottle_TM

full_join(metabar_detections_bottle_TM,qPCR_detections_bottle_TM) %>%
  filter(., !is.na(metabarcoding_PA)) %>%
    filter(., !is.na(qPCR_PA)) %>%
  mutate(., Detection = case_when(qPCR_PA ==0 & metabarcoding_PA ==0 ~"Both",
                                  qPCR_PA ==0 & metabarcoding_PA >0 ~"Metabarcoding Only",
                                  qPCR_PA >0 & metabarcoding_PA ==0 ~"qPCR Only",
                                  qPCR_PA >0 & metabarcoding_PA >0 ~"Both",
)) %>%
  filter(., !is.na(Detection)) %>%
  group_by(`Target Name`) %>%
```

```
  dplyr::count(Detection) -> detection_comparison_eDNA_bothc_TM
```

```
## Joining with `by = join_by(New_name_no_tech, `Target Name`)`
```

```
detection_comparison_eDNA_bothc_TM %>%
  group_by(`Target Name`) %>%
  mutate(., tot_obs = sum(n)) %>%
  mutate(., perc=n/tot_obs)
```

```
## # A tibble: 10 x 5
## # Groups:   Target Name [4]
##    `Target Name` Detection              n tot_obs   perc
##    <chr>         <chr>              <int>   <int>  <dbl>
##  1 Beluga        Both                  36      62 0.581
##  2 Beluga        Metabarcoding Only     3      62 0.0484
##  3 Beluga        qPCR Only             23      62 0.371
##  4 Chinook       Both                  56      62 0.903
##  5 Chinook       Metabarcoding Only     4      62 0.0645
##  6 Chinook       qPCR Only              2      62 0.0323
##  7 Coho          Both                  62      62 1
##  8 Eulachon      Both                  60      62 0.968
##  9 Eulachon      Metabarcoding Only     1      62 0.0161
## 10 Eulachon      qPCR Only              1      62 0.0161
```

**Twentymile River Visual and Molecular Beluga Detection Rates**

**Detection Rate Table**

| Method | Beluga Occurrence | Detection Rate |
|---|---|---|
| Visual Observation | Before | 0.000 |
| Visual Observation | During | 0.692 |
| qPCR | Before | 0.833 |
| qPCR | During | 0.643 |
| MetaBar | Before | 0.000 |
| MetaBar | During | 0.462 |

## Kenai River

**Beluga Observations: Visual vs. eDNA Metabarcoding vs. qPCR** eDNA and visual observations of Cook Inlet Beluga whales were largely concordant within Kenai River (Figure 6). These results demonstrate that both metabarcoding and qPCR eDNA approaches can provide valuable data on the occurrence of belugas in the absence of visual surveys which is particularly useful when visual observations are difficult to obtain (i.e. Winter). Here routine and consistent visual surveys were not conducted in Winter and thus robust occurrence data is missing from this time series. However, eDNA data had frequent detections of beluga in the Kenai River during the Winter, strongly suggesting that beluga utilize these habitats seasonally even when the River is completely frozen over.

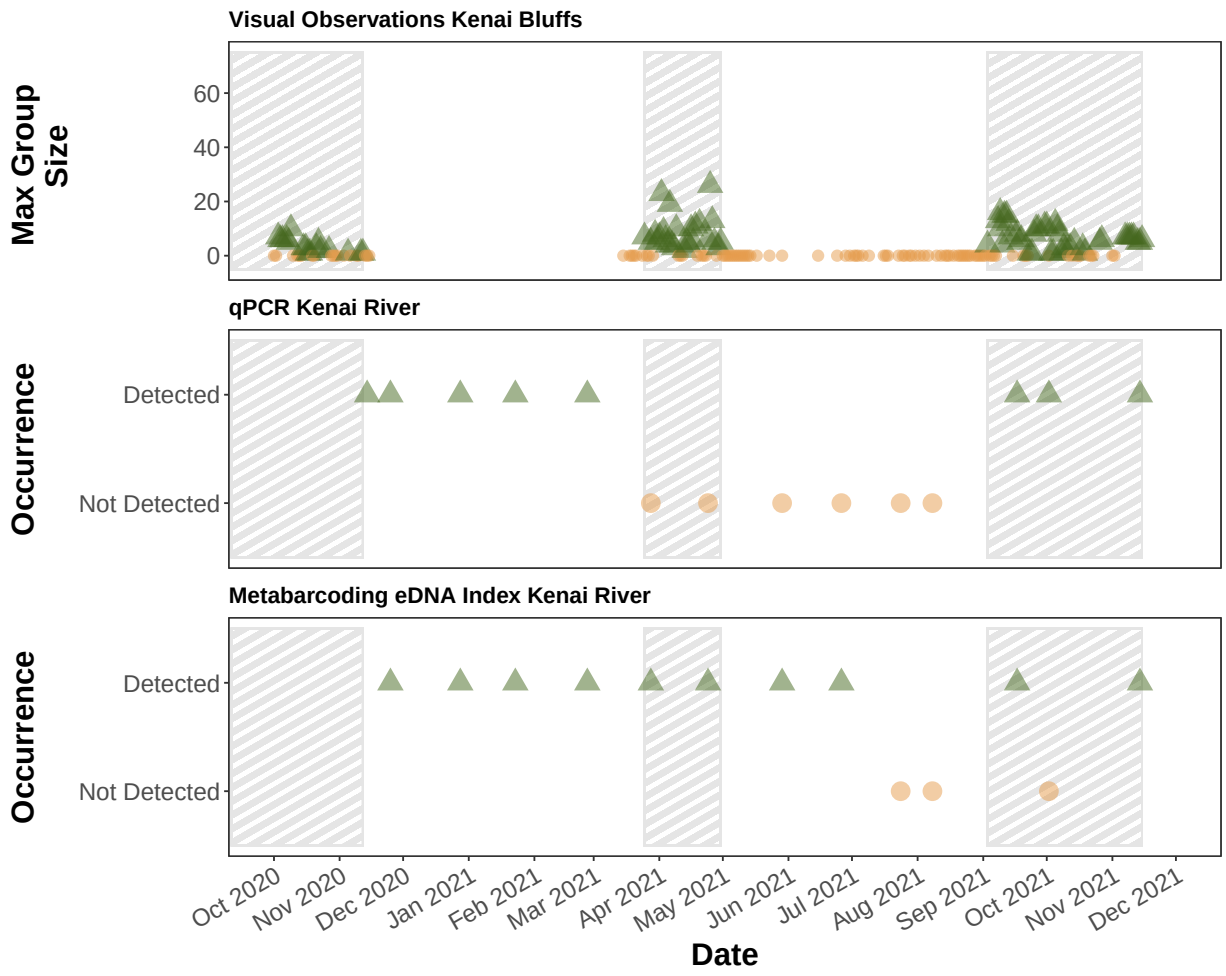
## Figure 5. Visual and eDNA detections of beluga occurrence in Kenai River

```
meta_data_only_qPCR_species_long %>%
  filter(., River %in% c("Kenai")) %>%
  filter(., Species=="Delphinapterus leucas") %>%
    group_by(Sample_date) %>%
  dplyr::summarise(tot_reads =sum(nReads)) %>%
  mutate(., PA = if_else(tot_reads > 0, "Detected","Not Detected")) -> beluga_intk2


beluga_intk2$PA <- factor(beluga_intk2$PA, levels = c("Not Detected", "Detected"))
beluga_intk2 %>%
  ggplot(aes(x=Sample_date, y=PA, color=PA,shape=PA)) +
  geom_rect_pattern(
    xmin = as.Date("2021-09-03"), xmax = as.Date("2021-11-15"), ymin = 0.5, ymax = 2.5,
    fill            = 'white',
    colour          = 'grey90',
    pattern_density = 0.35,
    pattern_fill    = 'grey90',
    pattern_colour  = 'grey90',
    pattern="stripe") +
  geom_rect_pattern(
    xmin = as.Date("2020-09-11"), xmax = as.Date("2020-11-12"), ymin = 0.5, ymax = 2.5,
    fill            = 'white',
    colour          = 'grey90',
    pattern_density = 0.35,
    pattern_fill    = 'grey90',
    pattern_colour  = 'grey90',
    pattern="stripe") +
  geom_rect_pattern(
    xmin = as.Date("2021-03-25"), xmax = as.Date("2021-04-30"), ymin = 0.5, ymax = 2.5,
    fill            = 'white',
    colour          = 'grey90',
    pattern_density = 0.35,
    pattern_fill    = 'grey90',
    pattern_colour  = 'grey90',
    pattern="stripe") + geom_point(alpha=0.5, size=5)   +
    scale_size(range = c(3,5)) +
  guides(color=guide_legend(override.aes = list(size = c( 2, 5))) ) +
  theme_bw()+
  ggtitle('Metabarcoding eDNA Index Kenai River') +  xlab("Date")+scale_x_date(limits = as.Date(c("2020-
panel.background = element_blank(), axis.text = element_text(size=14), axis.title = element_text(size=18
```

```
beluga_visual_3 %>%
  filter(., Site.Name=="Kenai River") %>%
    mutate(., PA = if_else(max_group_size > 0, "Detected","Not Detected"),
              PA_plot = if_else(max_group_size > 0, 5,2)) %>%
  ggplot(., aes(x=Sample_date,y=max_group_size, color=Were.belugas.observed.during.the.monitoring.sessio
  geom_rect_pattern(
    xmin = as.Date("2021-09-03"), xmax = as.Date("2021-11-15"), ymin = -5, ymax = 75,
    fill            = 'white',
```

```
    colour          = 'grey90',
    pattern_density = 0.35,
    pattern_fill    = 'grey90',
    pattern_colour  = 'grey90',
    pattern="stripe") +
  geom_rect_pattern(
    xmin = as.Date("2020-09-11"), xmax = as.Date("2020-11-12"), ymin = -5, ymax = 75,
    fill            = 'white',
    colour          = 'grey90',
    pattern_density = 0.35,
    pattern_fill    = 'grey90',
    pattern_colour  = 'grey90',
    pattern="stripe") +
  geom_rect_pattern(
    xmin = as.Date("2021-03-25"), xmax = as.Date("2021-04-30"), ymin = -5, ymax = 75,
    fill            = 'white',
    colour          = 'grey90',
    pattern_density = 0.35,
    pattern_fill    = 'grey90',
    pattern_colour  = 'grey90',
    pattern="stripe") + geom_point(alpha=0.5, aes(size=PA_plot))  +
    scale_size(range = c(3,5)) +
  guides(color=guide_legend(override.aes = list(size = c( 2, 5))) )  +
  theme_bw() +theme(axis.text.x = element_text(angle=30, hjust = 1))  + ylab("Max Group \n Size") + xlab
))  +scale_x_date(limits = as.Date(c("2020-10-01","2021-12-01")), breaks = scales::breaks_pretty(15))+
panel.background = element_blank(), axis.text = element_text(size=14), axis.title = element_text(size=18
axis.title.x  = element_blank(), axis.ticks.x = element_blank())-> visual_beluga_plot
```




**Kenai River Visual and Molecular Beluga Detection Rates**

**Fall**

**Table S4.**

| Method | Beluga Occurrence | Detection Rate |
|---|---|---|
| Visual Observation | Before | 0.0000000 |
| Visual Observation | During | 0.6721311 |
| qPCR | Before | 0.0000000 |
| qPCR | During | 1.0000000 |
| Metabarcoding | Before | 0.3333333 |
| Metabarcoding | During | 0.6666667 |

```
qPCR_4_species %>%
  mutate(., `Target Name`=if_else(`Target Name`=="cetacea","Beluga",`Target Name`)) %>%
  filter(., River %in% c("Kenai")) %>%
  filter(., `Target Name`=="Beluga") %>%
```

```
  group_by(Sample_date) %>%
    dplyr::summarise( avg_Detection_qPCR = mean(mean_dnacopy_L)) %>%
  mutate(., avg_Detection_qPCR= if_else(is.na(avg_Detection_qPCR),0,avg_Detection_qPCR))  %>%
    mutate(., Detection_qPCR= if_else(avg_Detection_qPCR>0,1,0)) -> kenai_detect_qPCR


combined_data_long_reads_c %>%
   filter(., River %in% c('Kenai')) %>%
  filter(., Species=="Delphinapterus leucas") %>%
  group_by(Sample_date) %>%
    mutate(., Detection_metabar = if_else(nReads>0,1,0)) %>%
  dplyr::summarise( avg_Detection_metabar = mean(Detection_metabar)) %>%
      mutate(., Detection_metabar= if_else(avg_Detection_metabar>0,1,0))  -> kenai_detect_metabar
```

Importantly,visual observations of Belugas in Kenai River demonstrate a noted absence in the summer with presence in Fall and Spring during AKBMP and Sonia Kumar's visual observations (Figure 6). Here we focus on what fish species were observed in higher abundances when Beluga were present by isolating the Winter to Spring transition and the Summer to Fall transition. In both cases Beluga are conspicuously absent from the river in Winter/Summer, but present in Spring (March 25-April 30th, 2021) and Fall (September 1st-November 2021).

**Spring**

**Table S5.**

| Method | Beluga Occurrence | Detection Rate |
|---|---|---|
| Visual Observation | After | 0.0000000 |
| Visual Observation | During | 0.7027027 |
| qPCR | After | 0.0000000 |
| qPCR | During | 0.0000000 |
| Metabarcoding | After | 0.5000000 |
| Metabarcoding | During | 1.0000000 |

## Beluga occurrence in Twentymile River correlated with Coho salmon run timing

We demonstrate that eDNA metabarcoding can successfully capture changes in fish assemblages on a sub-weekly time scale, allowing for comparison of Beluga habitat use against anadromous fish run timing. Specifically, we find that Beluga occurrence in Twentymile River was strongly correlated with increases in Coho abundances in summer/fall 2021 (Figure 4 & Figures S1-S3).

**20M qPCR Results**

```
natparks.pals("CraterLake", 7) -> cl
```

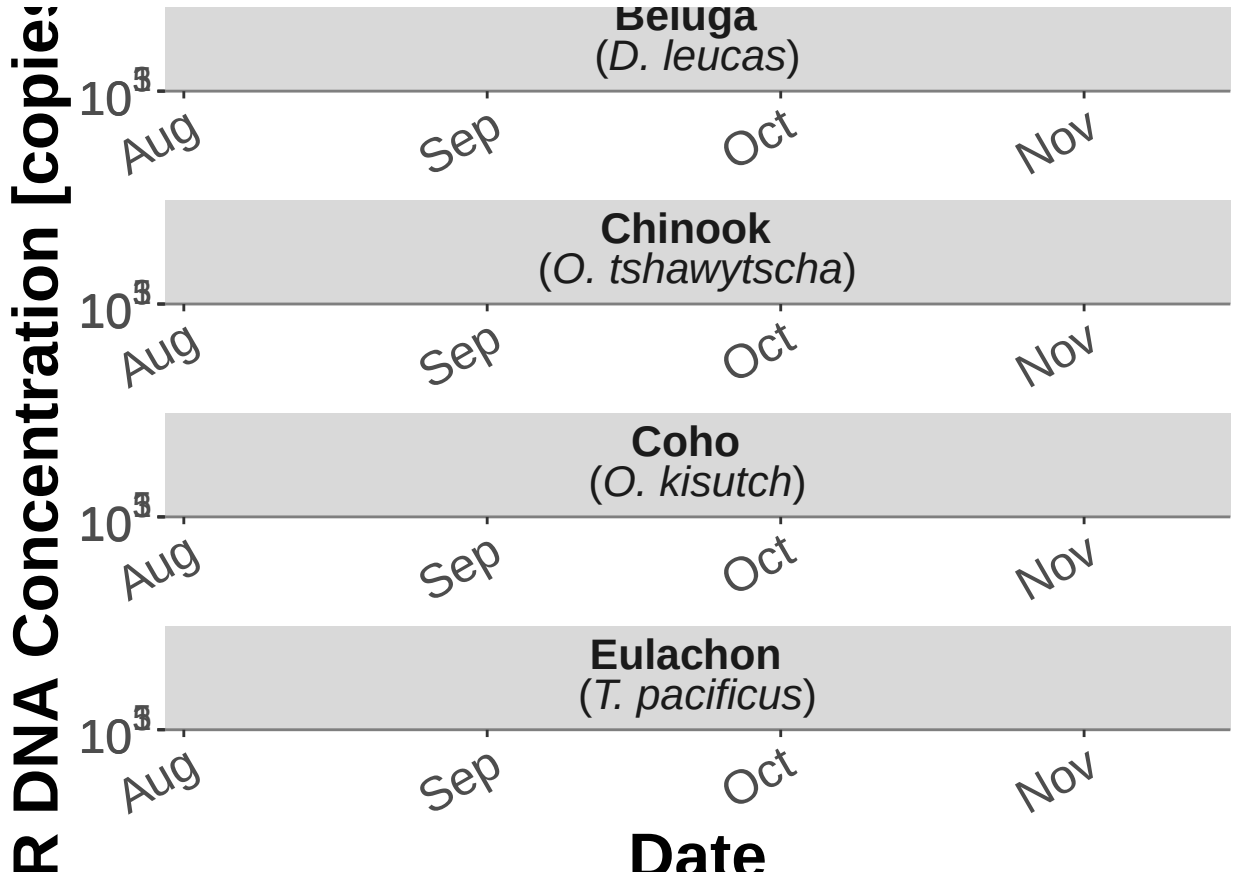
```
palette_cl <- c("grey50",cl[3],cl[1],cl[4])

qPCR_4_species %>%
  filter(., River=="Twentymile") %>%
     mutate(., `Target Name`=case_when(`Target Name`=="cetacea"~ "**Beluga** <br> (*D. leucas*)" ,
                                       `Target Name`=="Eulachon" ~ "**Eulachon** <br> (*T. pacificus*)",
                                       `Target Name`=="Chinook"~ "**Chinook** <br> (*O. tshawytscha*)",
                                        `Target Name`=="Coho" ~ "**Coho** <br> (*O. kisutch*)")) %>%
ggplot(aes(x=Sample_date, y=mean_dnacopy_L+1,color=`Target Name`)) + geom_point()+ facet_wrap(`Target Na
   theme(legend.position = "none",axis.text.x = element_text(angle=30, hjust = 1), panel.grid.major = el
panel.background = element_blank(), axis.text = element_text(size=18), axis.title = element_text(size=24
  scale_y_log10(labels = label_log(), n.breaks=5)
```

**Figure S6.1 20M River**

```
## Warning in scale_y_log10(labels = label_log(), n.breaks = 5): log-10
## transformation introduced infinite values.

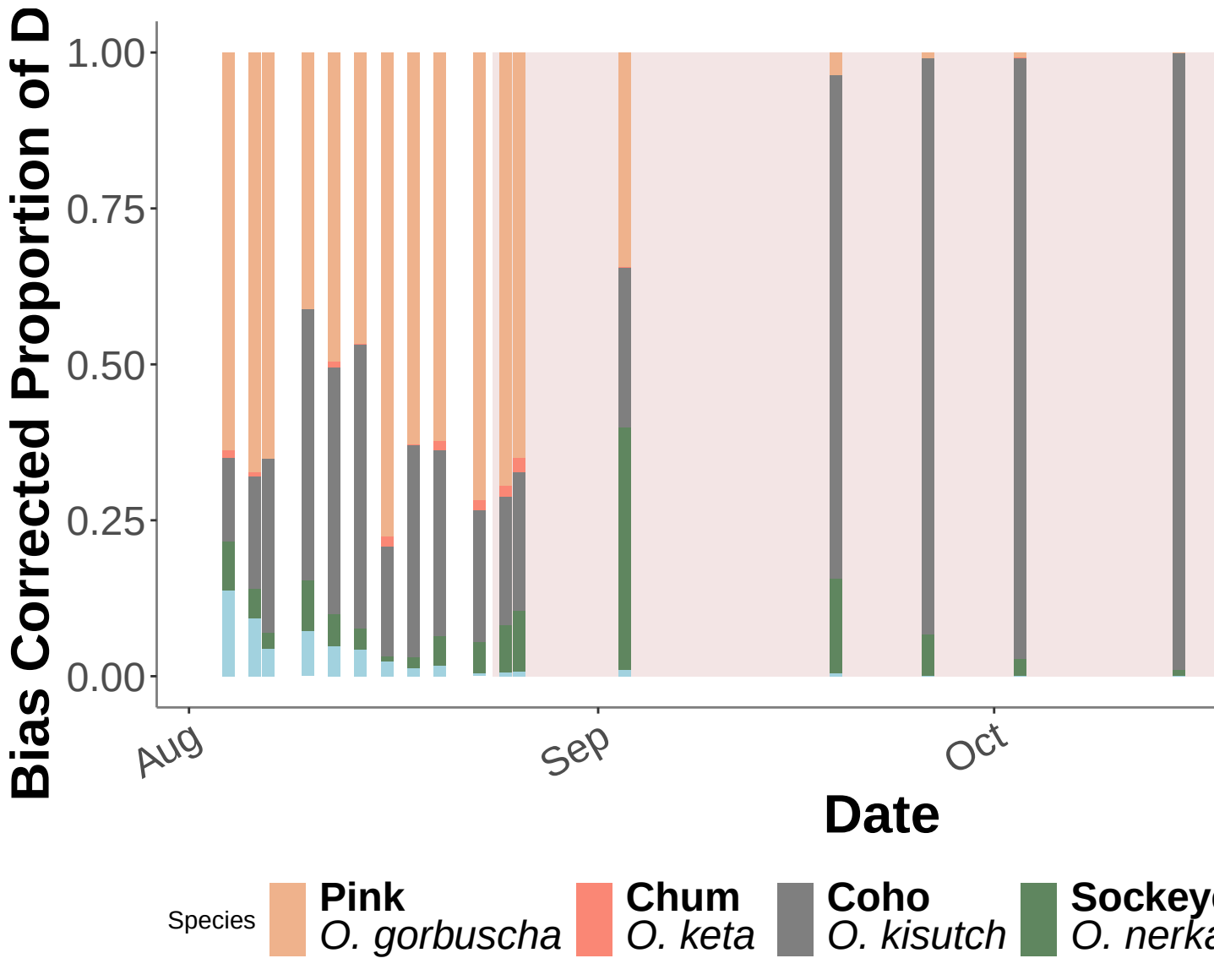
## Warning: Removed 21 rows containing missing values or values outside the scale range
## (`geom_point()`).
```




```
ggsave(file = here::here("analysis", "figures", "Figure_S6.1_20m_qPCR.png"),
 width = 12,
 height = 12
 )
```

```
## Warning in scale_y_log10(labels = label_log(), n.breaks = 5): log-10 transformation introduced infini
## Removed 21 rows containing missing values or values outside the scale range
## (‘geom_point()‘).
```

**Figure S6.2. 20M Amplification Bias Corrected Metabarcoding Results**




```
library(rcompanion)
```

```
## Registered S3 method overwritten by 'DescTools':
##   method        from
##   print.palette NatParksPalettes
```

```
Sum = groupwiseMedian(avgprop ~ time_chunk * species,
                      data=qm_bias_corrected)
```

```
Sum %>%
  dplyr::select(-n, -Conf.level) %>%
  pivot_wider(names_from=time_chunk, names_glue = "{time_chunk}_{.value}",
    values_from = c(Median, Bca.lower,Bca.upper)) %>%
  mutate(., Overlap = case_when( During_Bca.lower - Before_Bca.upper > 0 ~"During > Before",
                                 During_Bca.upper - Before_Bca.lower < 0 ~"Before > During",
                                 TRUE ~"Overlap")) %>%
  dplyr::select(species, Overlap) -> overlap

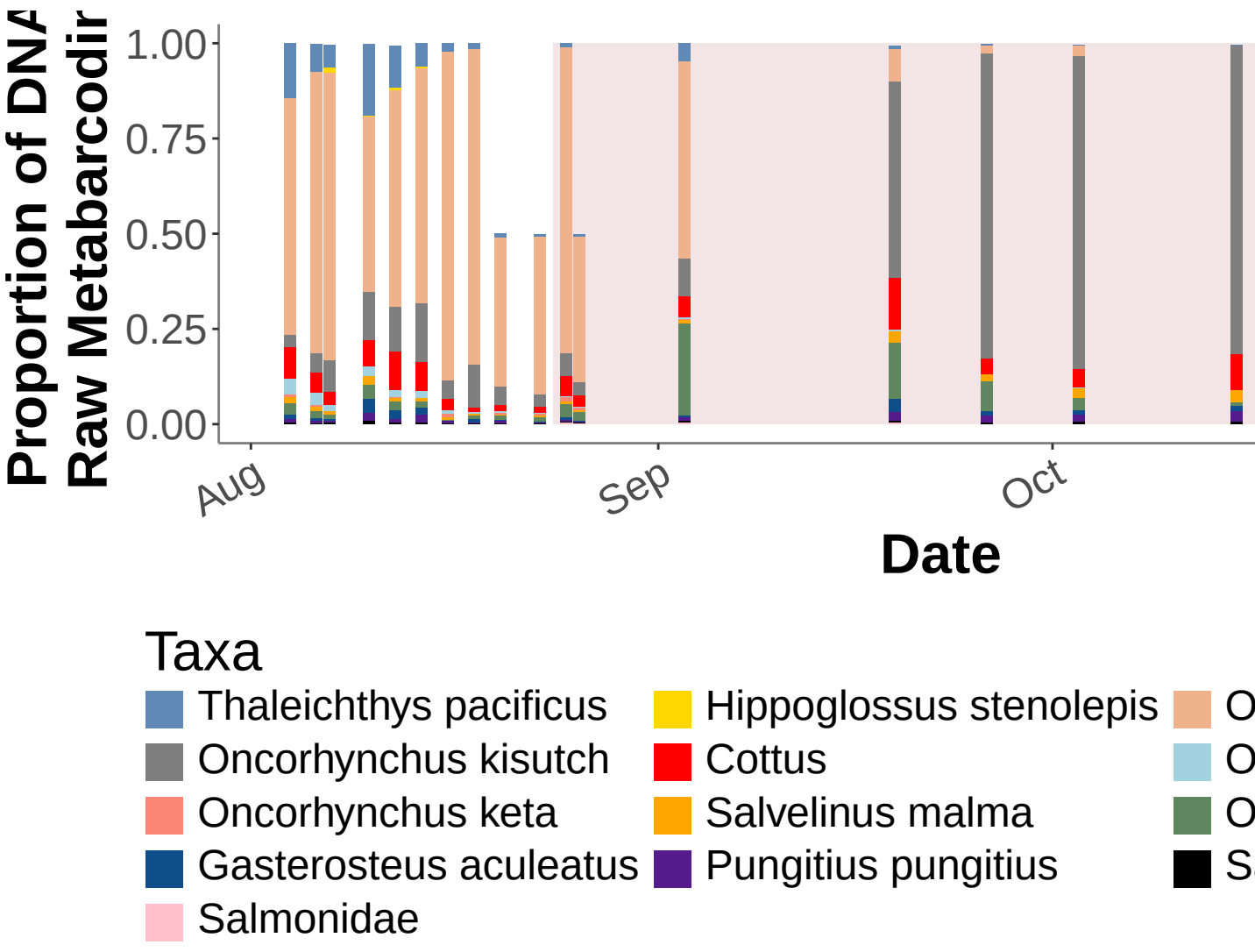
Sum %>%
    left_join(overlap) %>%
  dplyr::select(Period=time_chunk, Species=species, n, Median, `LCI-0.95` = Bca.lower, `UCI-0.95`=Bca.u
  arrange(Species) %>%
  mutate(., ) %>%
  kable()
```

```
## Joining with 'by = join_by(species)'
```

| Period | Species | n | Median | LCI-0.95 | UCI-0.95 | Confidence Interval |
|---|---|---|---|---|---|---|
| After | Oncorhynchus gorbuscha | 3 | 1.20e-02 | 1.07e-02 | 1.33e-02 | Overlap |
| Before | Oncorhynchus gorbuscha | 24 | 1.21e-01 | 1.12e-01 | 1.53e-01 | Overlap |
| During | Oncorhynchus gorbuscha | 32 | 2.06e-01 | 1.97e-02 | 2.15e-01 | Overlap |
| After | Oncorhynchus keta | 3 | 7.00e-07 | 7.00e-07 | 7.00e-07 | Overlap |
| Before | Oncorhynchus keta | 24 | 7.30e-04 | 1.90e-06 | 1.26e-03 | Overlap |
| During | Oncorhynchus keta | 32 | 8.20e-04 | 1.00e-06 | 5.00e-03 | Overlap |
| After | Oncorhynchus kisutch | 3 | 2.96e-01 | 2.94e-01 | 2.99e-01 | Overlap |
| Before | Oncorhynchus kisutch | 24 | 5.83e-02 | 2.93e-02 | 1.29e-01 | Overlap |
| During | Oncorhynchus kisutch | 32 | 9.84e-02 | 7.67e-02 | 1.92e-01 | Overlap |
| After | Oncorhynchus nerka | 3 | 2.18e-02 | 2.07e-02 | 2.29e-02 | Overlap |
| Before | Oncorhynchus nerka | 24 | 1.11e-02 | 9.56e-03 | 1.58e-02 | Overlap |
| During | Oncorhynchus nerka | 32 | 1.72e-02 | 9.63e-03 | 2.84e-02 | Overlap |
| After | Oncorhynchus tshawytscha | 3 | 3.32e-03 | 2.09e-03 | 3.32e-03 | Before > During |
| Before | Oncorhynchus tshawytscha | 24 | 1.73e-02 | 1.42e-02 | 2.15e-02 | Before > During |
| During | Oncorhynchus tshawytscha | 32 | 2.33e-03 | 1.03e-03 | 2.77e-03 | Before > During |

Coho and Sockeye corrected proportional abundances were signficantly higher during the occurrence of belugas in Twentymile River. Coho had the largest magnitude of change. Bias corrected metabarcoding data provides similar lines of evidence to quantitative absolute abundances.

**Figure S6.3. 20M Raw Metabarcoding Results**

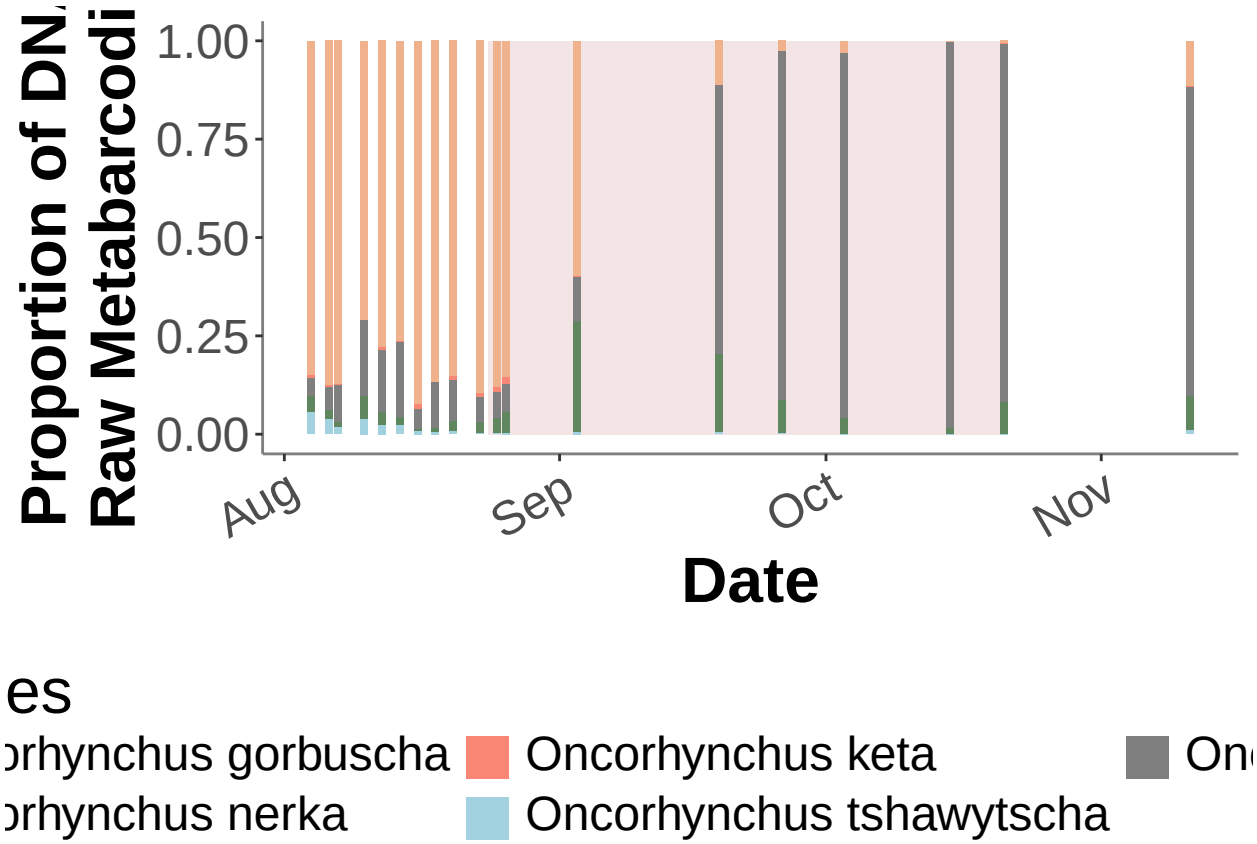



```
eDNA_index_20m_plotting %>%
      filter(., ID_mifish %in% c("Oncorhynchus gorbuscha" ,  "Oncorhynchus keta"    ,   "Oncorhynchus kis
                                 "Oncorhynchus tshawytscha")) %>%
  ungroup() %>%
  group_by(Seq_number) %>%
  mutate(., tot_reads2= sum(nReads),
         prop_reads2= nReads/tot_reads2) %>%
  group_by(ID_mifish,Sample_date) %>%
  dplyr::summarise(avgprop=mean(prop_reads2)) %>%
  ggplot(aes(x = Sample_date, fill = ID_mifish, y = avgprop)) +
  annotate("rect", xmin = as.Date("2021-08-24"), xmax = as.Date("2021-10-21"), ymin = 0, ymax = 1, alpha
  geom_col() +
  labs(x="Date", y="Proportion of DNA \n Raw Metabarcoding", fill="Species", color="") +
   theme(legend.position = "none",axis.text.x = element_text(angle=30, hjust = 1), panel.grid.major = el
```

```
panel.background = element_blank(), axis.text = element_text(size=18), axis.title = element_text(size=24
    theme(legend.text = element_markdown(size = 18), legend.position="bottom") +
  scale_fill_manual(values = c("#efb28c","#fa8775","grey50","#5f865f","#a2d2df"))  +
  guides(fill = guide_legend(nrow = 2, byrow = TRUE, title.position = "top"))
```

**Figure S6.4 20M Raw Metabarcoding Salmon**

```
## `summarise()` has regrouped the output.
## Ignoring unknown labels: * colour : ""
## i Summaries were computed grouped by ID_mifish and Sample_date.
## i Output is grouped by ID_mifish.
## i Use `summarise(.groups = "drop_last")` to silence this message.
## i Use `summarise(.by = c(ID_mifish, Sample_date))` for per-operation grouping
##   (`?dplyr::dplyr_by`) instead.
```




```
ggsave(file = here::here("analysis", "figures", "Figure_S6.4_20M_metabarcoding_IDs_salmon.png"),
  width = 12,
  height = 8
  )
```

```
## Ignoring unknown labels:
## * colour : ""
```

From the most abundantly detected taxa (over 1,000 reads detected in at least 10 technical replicates), we found significant differences in eDNA index scores before and during beluga occurrences in Twentymile River.

```
##  [1] "Thaleichthys pacificus"   "Hippoglossus stenolepis"
##  [3] "Oncorhynchus gorbuscha"   "Oncorhynchus kisutch"
##  [5] ""                         "Oncorhynchus tshawytscha"
##  [7] "Oncorhynchus keta"        "Salvelinus malma"
##  [9] "Oncorhynchus nerka"       "Gasterosteus aculeatus"
## [11] "Pungitius pungitius"      "Salvelinus alpinus"
```

Here we see Coho, Sockeye, and 9 Spined Stickleback had higher eDNA index scores during beluga occurrences. Coho had the largest change in magnitude, however from eDNA index scores we have no ability to compare between species and thus understand which species were the most abundant in Twentymile River.

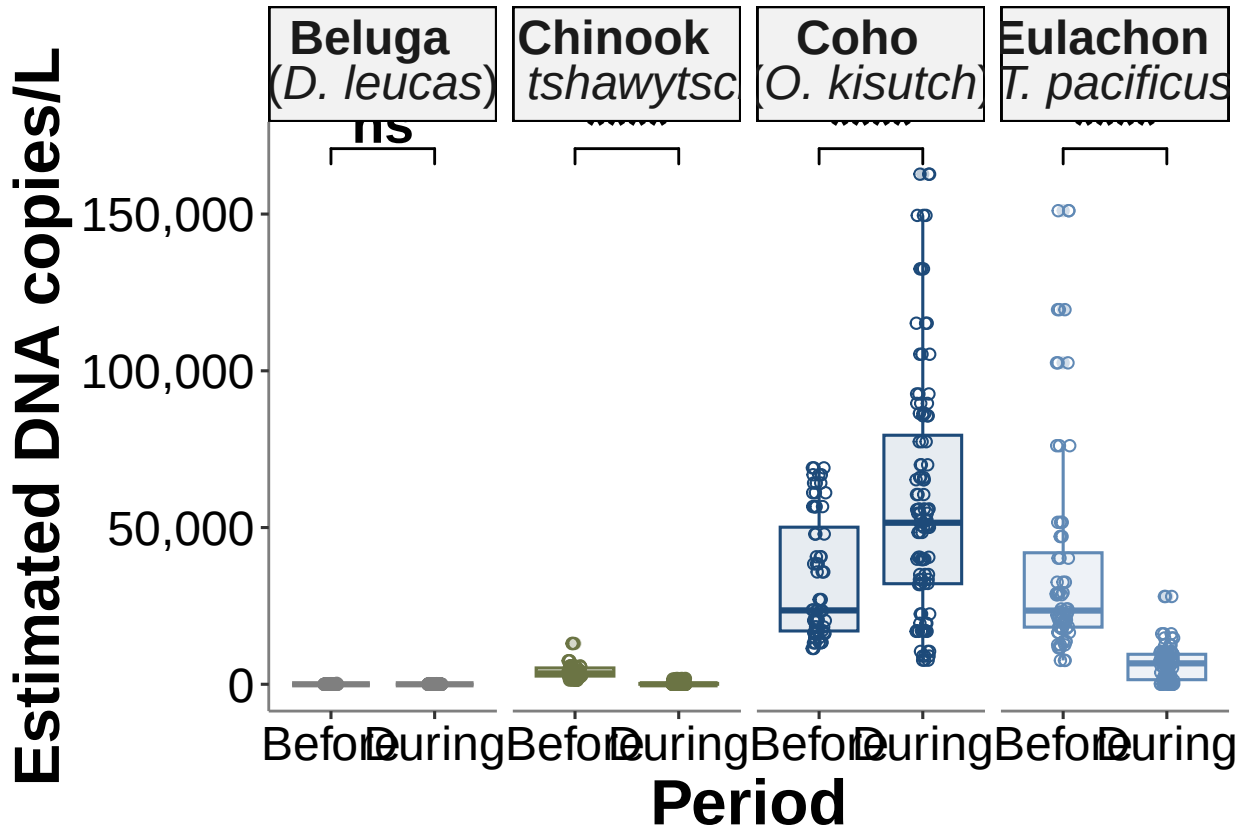
```
  my_comparisons <- list( c("Before", "During"))

qPCR_4_species %>%
  mutate(time_chunk = case_when(Sample_date < '2021-08-16' ~"Before",
    Sample_date > "2021-10-21" ~"After",
    TRUE ~"During")) %>%
  filter(., time_chunk != "After") %>%
 mutate(., Units ="copies/L") %>%
  filter(., River=="Twentymile") %>%
    mutate(., species=case_when(`Target Name`=="cetacea"~ "**Beluga** <br> (*D. leucas*)" ,
                                `Target Name`=="Eulachon" ~ "**Eulachon** <br> (*T. pacificus*)",
                                `Target Name`=="Chinook"~ "**Chinook** <br> (*O. tshawytscha*)",
                                 `Target Name`=="Coho" ~ "**Coho** <br> (*O. kisutch*)")) %>%
    filter(., !is.na(mean_dnacopy_L)) %>%
  ggplot(aes(x = time_chunk, colour = species, y = mean_dnacopy_L)) +
  geom_boxplot(aes(fill=species), alpha=0.1) +
  geom_point(position = position_jitterdodge(jitter.width = 0.2),shape = 1) +scale_color_manual(values =
    labs(x="Period", y="Estimated DNA copies/L") +
  theme_pubr()  +theme(legend.position = "none",
strip.text = element_markdown(size = 18, face = "plain"), panel.grid.major = element_blank(),panel.grid
panel.background = element_blank(), axis.text = element_text(size=18), axis.title = element_text(size=24
  stat_compare_means(comparisons = my_comparisons, label = "p.signif", size = 7, color="black", fontfac
  scale_y_continuous(labels = comma) -> Figure_S6.5

  Figure_S6.5$layers[[3]]$aes_params$fontface <- "bold"

  Figure_S6.5
```

**Figure S6.5 20M qPCR Abundance Differences Before and During Beluga Occupancy**




```
ggsave(file = here::here("analysis", "figures", "Figure_S6.5_20m_qPCR_diff_abund.png"),
 width = 12,
 height = 12
 )
```

**Twentymile Summary** Coho and Sockeye are known prey items for Cook Inlet Belugas (Quakenbush et al. 2015). Traditional ecological knowledge from informal interviews with local fisherman additionally supported that the timing of the Coho run coincided with the timing of Belugas arriving in Twentymile River both in 2021 and previous years for decades. Thus our findings of beluga occurrence being strongly correlated with Coho run timing aligns well with previous research (McGuire et al. 2020; Nelson et al. 2018; Castellote et al. 2020). We note that multiple lines of evidence suggest Coho DNA was far more abundant across all time periods in Twentymile River as demonstrated by quantitative metabarcoding, bias corrected metabarcoding, qPCR, and eDNA index scores. Together, these results suggest that Cook Inlet Beluga utilization of Twentymile River in Fall is associated with the Coho run. Most importantly, these results demonstrate that weekly eDNA sampling efforts can provide sufficient resolution to capture changes in anadromous fish abundances and allow for co-occurrence analyses to inform habitat use of Cook Inlet Beluga whales.

## Temporal and Spatial Variation in Kenai River

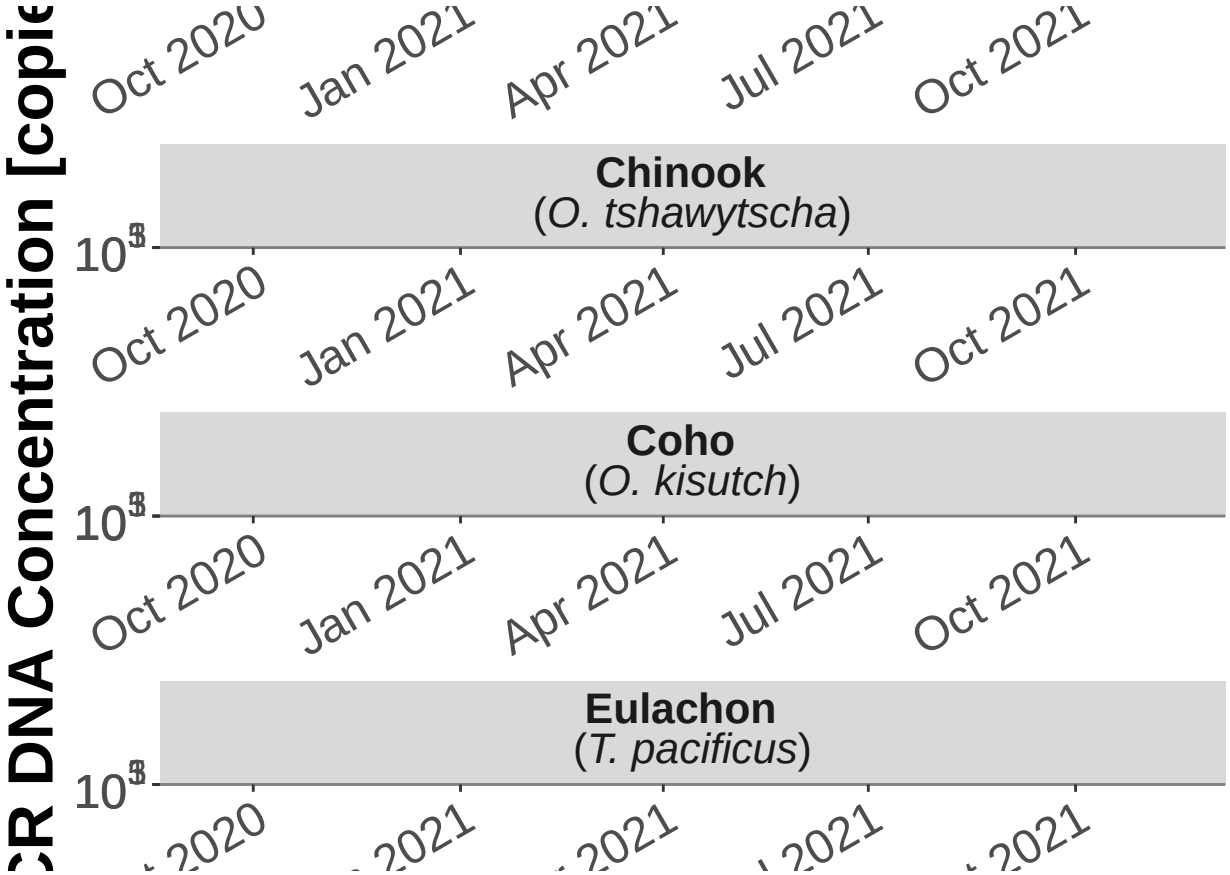
```
natparks.pals("CraterLake", 7) -> cl

palette_cl <- c("grey50",cl[3],cl[1],cl[4])

qPCR_4_species %>%
  filter(., River=="Kenai") %>%
     mutate(., `Target Name`=case_when(`Target Name`=="cetacea"~ "**Beluga** <br> (*D. leucas*)" ,
                                       `Target Name`=="Eulachon" ~ "**Eulachon** <br> (*T. pacificus*)",
                                       `Target Name`=="Chinook"~ "**Chinook** <br> (*O. tshawytscha*)",
                                        `Target Name`=="Coho" ~ "**Coho** <br> (*O. kisutch*)")) %>%
ggplot(aes(x=Sample_date, y=mean_dnacopy_L+1,color=`Target Name`)) + annotate("rect",
    xmin = as.Date("2021-09-03"), xmax = as.Date("2021-11-15"), ymin = 0.5, ymax = 1000000,
    alpha = .1,fill ="darkred") +
  annotate("rect",
    xmin = as.Date("2020-09-11"), xmax = as.Date("2020-11-12"), ymin = 0.5, ymax = 1000000,
   alpha = .1,fill ="darkred") +
  annotate("rect",
    xmin = as.Date("2021-03-25"), xmax = as.Date("2021-04-30"), ymin = 0.5, ymax = 1000000,
    alpha = .1,fill ="darkred")   +geom_point()+ facet_wrap(`Target Name`~., scale="free", ncol=1) +
   theme(legend.position = "none",axis.text.x = element_text(angle=30, hjust = 1), panel.grid.major = el
panel.background = element_blank(), axis.text = element_text(size=18), axis.title = element_text(size=24
  scale_y_log10(labels = label_log(), n.breaks=5)
```

**Figure S6.6 Kenai River**

```
## Warning: Removed 97 rows containing missing values or values outside the scale range
## (`geom_point()`).
```




```
ggsave(file = here::here("analysis", "figures", "Figure_S6.6_kenai_qPCR.png"),
 width = 12,
 height = 12
 )
```

```
## Warning: Removed 97 rows containing missing values or values outside the scale range
## (‘geom_point()‘).
```

We found higher Coho DNA concentrations during the Beluga occurrence period in Twentymile River in contrast to lower Chinook and Eulachon.

**Figure 6. Quantitative Metabarcoding Abundance Results with Coho Anchor**

```
combo_4_species_20m %>%
  mutate(time_chunk = case_when(Sample_date < '2021-08-16' ~"Before",
    Sample_date > "2021-10-21" ~"After",
    TRUE ~"During")) %>%
  filter(., time_chunk != "After") %>%
 mutate(., Units ="copies/L")-> qPCR_20m_analysis
```

```
library(rcompanion)
library(knitr)
Sum = groupwiseMedian(mean_est_dna ~ time_chunk * species,
                      data=qPCR_20m_analysis)


Sum %>%
  dplyr::select(-n, -Conf.level) %>%
  pivot_wider(names_from=time_chunk, names_glue = "{time_chunk}_{.value}",
    values_from = c(Median, Bca.lower,Bca.upper)) %>%
  mutate(., Overlap = case_when( During_Bca.lower - Before_Bca.upper > 0 ~"During > Before",
                                 During_Bca.upper - Before_Bca.lower < 0 ~"Before > During",
                                 TRUE ~"Overlap")) %>%
  dplyr::select(species, Overlap) -> overlap

Sum %>%
    left_join(overlap) %>%
  dplyr::select(Period=time_chunk, Species=species, n, Median, `LCI-0.95` = Bca.lower, `UCI-0.95`=Bca.u
  arrange(Species) %>%
  mutate(., ) %>%
  kable()
```

**Table**

```
## Joining with 'by = join_by(species)'
```

| Period | Species | n | Median | LCI-0.95 | UCI-0.95 | Confidence Interval |
|---|---|---|---|---|---|---|
| Before | Delphinapterus leucas | 14 | 21.8 | 11.5 | 71.1 | Overlap |
| During | Delphinapterus leucas | 20 | 157.0 | 37.7 | 321.0 | Overlap |
| Before | Oncorhynchus kisutch | 24 | 22400.0 | 17000.0 | 37500.0 | During > Before |
| During | Oncorhynchus kisutch | 32 | 51000.0 | 38200.0 | 64600.0 | During > Before |
| Before | Oncorhynchus tshawytscha | 24 | 3710.0 | 2850.0 | 4420.0 | Before > During |
| During | Oncorhynchus tshawytscha | 28 | 344.0 | 188.0 | 511.0 | Before > During |
| Before | Thaleichthys pacificus | 24 | 22600.0 | 18300.0 | 29600.0 | Before > During |
| During | Thaleichthys pacificus | 35 | 5960.0 | 1710.0 | 7710.0 | Before > During |

| Period | Species | n | Median | LCI-0.95 | UCI-0.95 | Confidence Interval |
|---|---|---|---|---|---|---|
| Before | Oncorhynchus gorbuscha | 24 | 57400 | 48700.00 | 69100 | Overlap |
| During | Oncorhynchus gorbuscha | 32 | 80300 | 1290.00 | 130000 | Overlap |
| Before | Oncorhynchus keta | 24 | 699 | 8.34 | 1140 | Overlap |
| During | Oncorhynchus keta | 32 | 1200 | 78.40 | 3320 | Overlap |
| Before | Oncorhynchus kisutch | 24 | 22400 | 17000.00 | 38300 | During > Before |
| During | Oncorhynchus kisutch | 32 | 51000 | 38500.00 | 64600 | During > Before |
| Before | Oncorhynchus nerka | 24 | 5340 | 3840.00 | 6390 | Overlap |
| During | Oncorhynchus nerka | 32 | 6170 | 2830.00 | 9170 | Overlap |
| Before | Oncorhynchus tshawytscha | 24 | 8590 | 7020.00 | 11100 | Before > During |
| During | Oncorhynchus tshawytscha | 32 | 1240 | 271.00 | 2080 | Before > During |

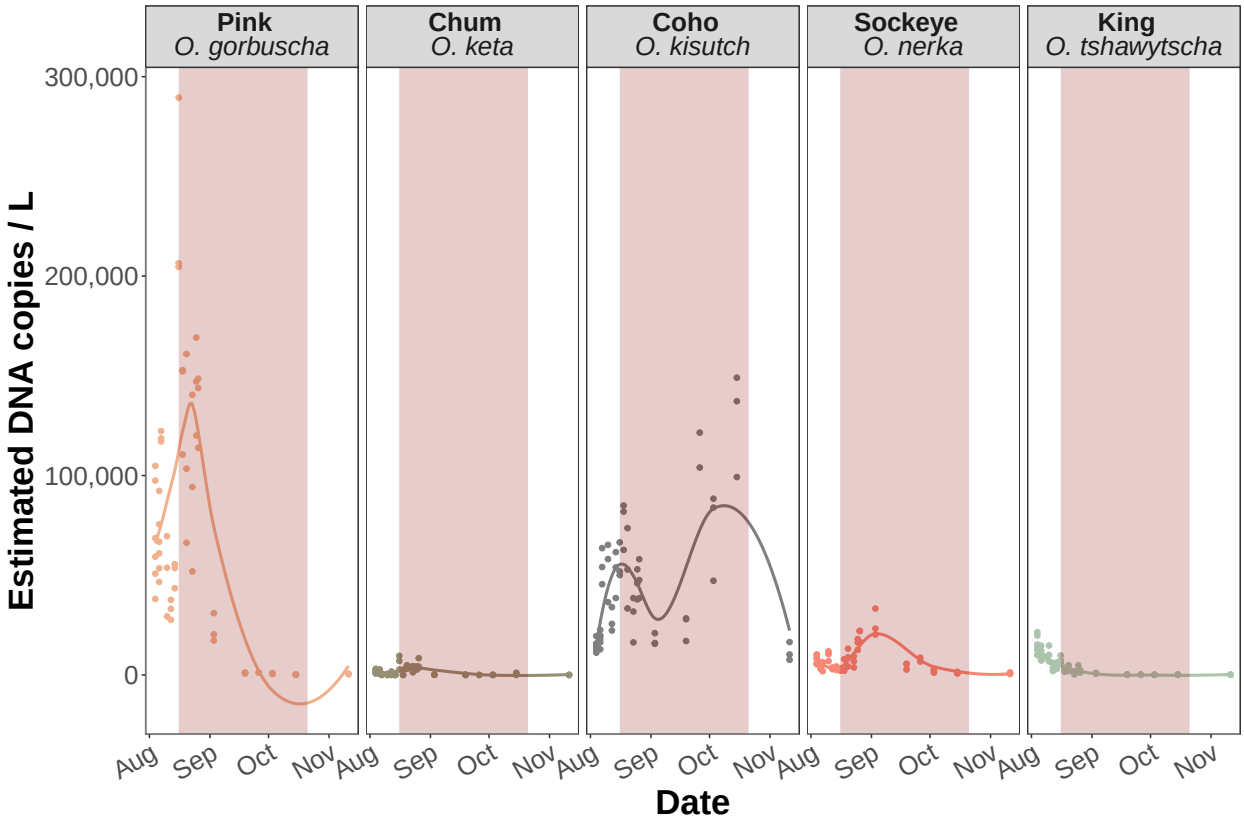
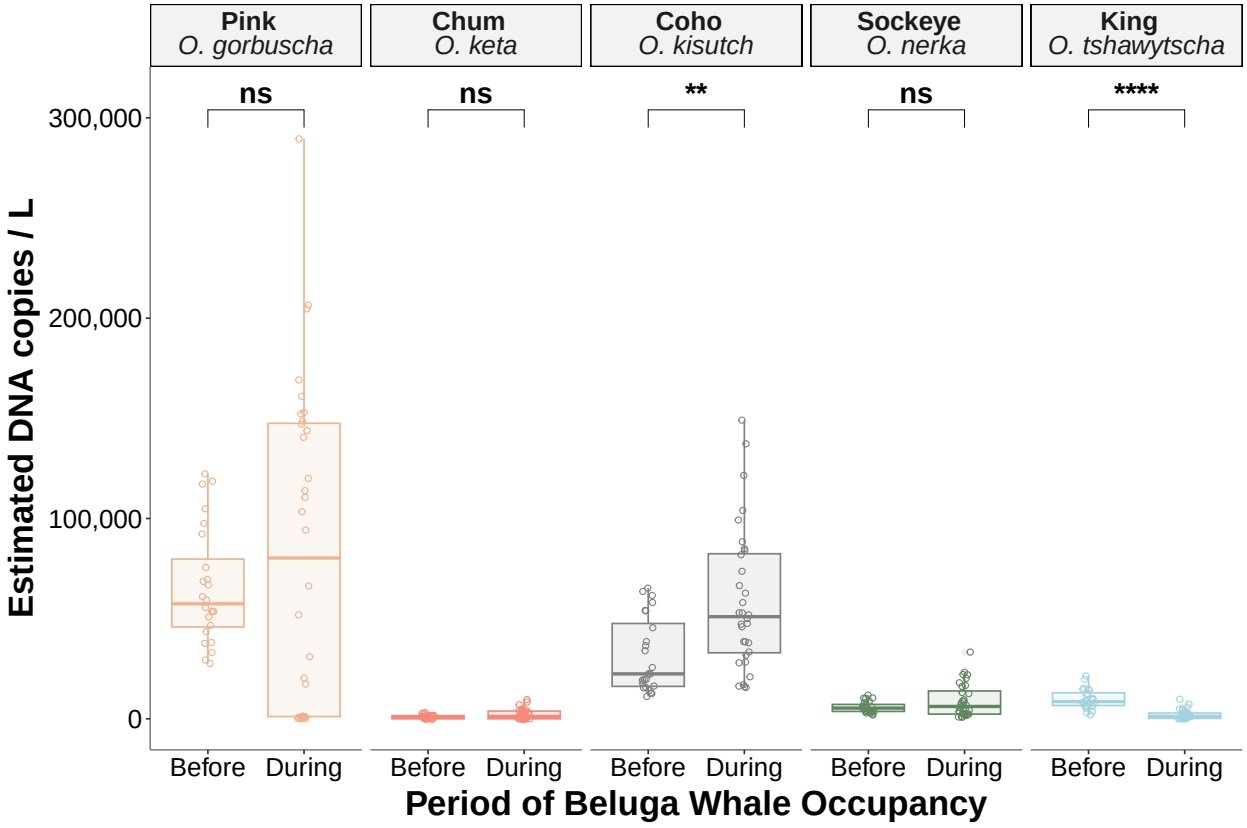

Pink
O. gorbuscha
Chum
O. keta
Coho
O. kisutch
Sockeye
O. nerka
King
O. tshawytscha
Estimated DNA copies / L
300,000
200,000
100,000
0
Aug
Sep
Oct
Nov
Date

**Figure S3**

**Figure 6.**




From the quantitative metabarcoding results, we found higher Coho DNA concentrations during Beluga occurrence in Twentymile River. The results align with traditional ecological knowledge of beluga behavior (Crews personal communication) and provide evidence that Beluga are swimming up Twentymile River to forage for Coho, a known prey target of Cook Inlet Belugas (Quakenbush et al. 2015).

The goal of these analyses is to understand the relative contribution of sampling date, site, and depth on the variance of fish communities. These results can inform sampling designs of future Cook Inlet eDNA applications.

### Phyloseq Reads

### Summary of what we found so far

Looks like the 2 sets of samples cluster closely to eachother. BUT for some reason the August bridge samples have a lot of O. nerka (all on run 1) whereas the dock samples have a lot of O. gorbuscha (all on run2). So the question is - do we think this is real, or do we have reason to think that this is the result of a lab effect (mislabel the site or extractions?, run effect?).

For Run 2 - the samples were more spread out and still cluster closely. So unlikely a pipetting error or anything of that sort. Samples extracted at the same time look different so does not suggest a batch effect.

Maybe just is that different. Both samples are quite swamped (>95% for a single species)

All the bridge samples had 2 filters and had extractions pooled... Is that the source of the isse?

qPCR data had differences in Chinook concentrations

**PERMANOVA**

```
# calculate distances
aitchison_dists <- physeq_obj_kenai_reads %>%
  tax_transform("identity", rank = "ID_mifish") %>%
  dist_calc("aitchison")

# the more permutations you request, the longer it takes
# but also the more stable and precise your p-values become
aitchison_perm <- aitchison_dists %>%
  dist_permanova(
    seed = 1234, # for set.seed to ensure reproducibility of random process
    n_processes = 1, n_perms = 999, # you should use at least 999!
    variables = c("Sample_date","Site","Bio_rep","Depth"),
    interactions= "Sample_date*Site*Bio_rep*Depth",
    by="terms"
  )
```

```
## 2026-11-10 00:23:46.552429 - Starting PERMANOVA with 999 perms with 1 processes
```

```
## 2026-11-10 00:24:07.09587 - Finished PERMANOVA
```

```
aitchison_perm
```

```
## psExtra object - a phyloseq object with extra slots:
##
## phyloseq-class experiment-level object
## otu_table()   OTU Table:         [ 39 taxa and 120 samples ]
## sample_data() Sample Data:       [ 120 samples by 10 sample variables ]
## tax_table()   Taxonomy Table:    [ 39 taxa by 10 taxonomic ranks ]
##
## psExtra info:
## tax_agg = "ID_mifish" tax_trans = "identity"
##
## aitchison distance matrix of size 120
## 18.13457 17.87039 29.61585 26.07153 27.05813 ...
##
## permanova:
## Permutation test for adonis under reduced model
## Terms added sequentially (first to last)
## Permutation: free
## Number of permutations: 999
##
## vegan::adonis2(formula = formula, data = metadata, permutations = n_perms, by = by, parallel = paral
##                                 Df SumOfSqs      R2       F Pr(>F)
## Sample_date                     12  16041.0 0.53220 15.1167  0.001 ***
## Site                             1   1062.6 0.03526 12.0169  0.001 ***
## Bio_rep                          1    109.8 0.00364  1.2412  0.247
## Depth                            1    147.4 0.00489  1.6668  0.080 .
## Sample_date:Site                 8   1658.3 0.05502  2.3441  0.001 ***
## Sample_date:Bio_rep             12   1376.7 0.04567  1.2973  0.029 *
## Site:Bio_rep                     1    154.1 0.00511  1.7430  0.071 .
```

```
## Sample_date:Depth               10   1896.6 0.06292  2.1447  0.001 ***
## Site:Depth                       1    186.5 0.00619  2.1085  0.035 *
## Bio_rep:Depth                    1    104.3 0.00346  1.1793  0.281
## Sample_date:Site:Bio_rep         5    598.2 0.01985  1.3529  0.064 .
## Sample_date:Site:Depth           4   1031.7 0.03423  2.9167  0.001 ***
## Sample_date:Bio_rep:Depth        8    852.3 0.02828  1.2048  0.112
## Site:Bio_rep:Depth               1    127.5 0.00423  1.4421  0.133
## Sample_date:Site:Bio_rep:Depth   3    372.7 0.01236  1.4048  0.083 .
## Residual                        50   4421.4 0.14669
## Total                          119  30141.0 1.00000
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

**Homogeneity of Dispersions test**

**Homogeneity of Dispersions test: Sample_date**

```
bd1$Sample_date$tukeyHSD$group %>%
  as.tibble(rownames="Comparisons") %>%
  arrange(`p adj`)
```

```
## # A tibble: 78 x 5
##    Comparisons            diff    lwr   upr      `p adj`
##    <chr>                 <dbl>  <dbl> <dbl>        <dbl>
##  1 2021-08-08-2021-04-24 -7.25 -10.8  -3.71 0.0000000235
##  2 2021-09-17-2021-04-24 -4.53  -7.05 -2.01 0.00000135
##  3 2021-08-08-2021-05-29 -6.04  -9.42 -2.65 0.00000172
##  4 2021-04-24-2020-12-28  5.33   2.21  8.45 0.00000531
##  5 2021-11-14-2021-04-24 -6.65 -10.7  -2.57 0.0000178
##  6 2021-07-24-2021-04-24 -3.94  -6.64 -1.24 0.000204
##  7 2021-09-17-2021-05-29 -3.32  -5.62 -1.01 0.000267
##  8 2021-05-29-2020-12-28  4.12   1.17  7.07 0.000474
##  9 2021-08-08-2020-11-25 -5.29  -9.11 -1.47 0.000570
## 10 2021-11-14-2021-05-29 -5.44  -9.39 -1.48 0.000636
## # i 68 more rows
```

**Homogeneity of Dispersions test: Site**

```
## # A tibble: 1 x 5
##   Comparisons  diff   lwr    upr `p adj`
##   <chr>       <dbl> <dbl>  <dbl>   <dbl>
## 1 Dock-Bridge -1.24 -2.34 -0.137  0.0279
```

We find a significant difference in eDNA signatures between temporal and spatial patterns, with date sampled explaning 61.6% variance while site samples explained only 4% variance. However, sampling data failed the homogeneity of dispersions test, indicating that the variance in community composition is signficantly different across the year. August, June, and July months had far lower diveristy and a greater proportion of total reads going to a smaller number of species, particularly Sockeye salmon which has a very large annual anadromous migration during this window. We did not find a signficant difference between samples collected at the surface and at depth. Interactive effects of sampling date with both site and depth were signficant and explained 1% or less of the total variance.

These results found temporal differences across the year are greater than site differences in tidally influenced systems with a high occurrence of anadromous fishes.

## Figure 7. Effects of Swamping on Non-Detections

Here we provide concrete evidence for swamping in metabarcoding results which is informative for future applications of quantitative metabarcoding approaches and our understanding of eDNA metabarcoding detection probabilities. These results support the quantitative metabarcoding framework developed by Shelton et al. 2023 (Shelton et al. 2023) and Gold et al. 2023 (Gold et al. 2023).

### Pre filtering

```
# Read metadata files for each sequence run and location
input_meta_path <- here("data","Metadata","metadata_seqrun1_20211225.csv")
metadata <- read.table(input_meta_path, header = 1, sep = ",", stringsAsFactors = F)

input_meta_path2 <- here("data","Metadata","metadata_seqrun2_20220309.csv")
metadata2 <- read.table(input_meta_path2, header = 1, sep = ",", stringsAsFactors = F)

input_meta_path_kenai_1 <- here("data","Metadata","metadata_kenai1_20220401.csv")
metadata_kenai_1 <- read.table(input_meta_path_kenai_1, header = 1, sep = ",", stringsAsFactors = F) %>%
  mutate(., Site_code=as.character(Site_code))

input_meta_path_kenai_2 <- here("data","Metadata","metadata_kenai2_20220412.csv")
metadata_kenai_2 <- read.table(input_meta_path_kenai_2, header = 1, sep = ",", stringsAsFactors = F) %>%
  mutate(., Site_code=as.character(Site_code))


input_meta_path_kenai_3 <- here("data","Metadata","metadata_kenai_run_3_20220518.csv")
metadata_kenai_3 <- read.table(input_meta_path_kenai_3, header = 1, sep = ",", stringsAsFactors = F) %>%
  mutate(., Site_code=as.character(Site_code))

bind_rows(metadata_kenai_3,metadata_kenai_2,metadata_kenai_1,metadata,metadata2) -> combined_metadata

combined_metadata$New_name %>% unique() %>%  length()
```

```
## [1] 501
```

```
combined_metadata$New_name_no_tech %>% unique() %>%  length()
```

```
## [1] 395
```

```
#pull in each ASV nested and explore ASVs and prior to filtering
kenai_1_asv_nested <- readRDS(file=here("data","eDNA_data","data","Anacapa_processed_sequence_data","ke
kenai_1_hash <- readRDS(file=here("data","eDNA_data","data","Anacapa_processed_sequence_data","kenai","

kenai_1_asv_nested$Step.4.low.read.depth[[1]] %>%
  left_join(kenai_1_hash) %>%
  group_by(across(c(-seq_number,-nReads))) %>%
```

```
  dplyr::summarise(nReads=sum(nReads)) %>%
  ungroup() %>%
  dplyr::select(-unique_sample, -barcode_1_col, -Miseq_run) %>%
  pivot_wider(names_from=sample, values_from = nReads, values_fill = 0) %>%
  pivot_longer(!sum.taxonomy, names_to = "sample", values_to= "nReads") %>%
  left_join(metadata_kenai_1, by=c("sample"="New_name"))-> kenai_1_asv_pre
```

```
## Joining with `by = join_by(seq_number, sum.taxonomy)`
## `summarise()` has regrouped the output.
```

```
kenai_2_asv_nested <- readRDS(file=here("data","eDNA_data","data","Anacapa_processed_sequence_data","ke
kenai_2_hash <- readRDS(file=here("data","eDNA_data","data","Anacapa_processed_sequence_data","kenai","

kenai_2_asv_nested$Step.4.low.read.depth[[1]] %>%
  left_join(kenai_2_hash) %>%
  group_by(across(c(-seq_number,-nReads))) %>%
  dplyr::summarise(nReads=sum(nReads)) %>%
  ungroup() %>%
  dplyr::select(-unique_sample, -barcode_1_col, -Miseq_run) %>%
  pivot_wider(names_from=sample, values_from = nReads, values_fill = 0) %>%
  pivot_longer(!sum.taxonomy, names_to = "sample", values_to= "nReads") %>%
   left_join(metadata_kenai_2, by=c("sample"="New_name"))-> kenai_2_asv_pre
```

```
## Joining with `by = join_by(seq_number, sum.taxonomy)`
## `summarise()` has regrouped the output.
```

```
kenai_3_asv_nested <- readRDS(file=here("data","eDNA_data","data","Anacapa_processed_sequence_data","ke

kenai_3_hash <- readRDS(file=here("data","eDNA_data","data","Anacapa_processed_sequence_data","kenai","

kenai_3_asv_nested$Step.4.low.read.depth[[1]] %>%
  left_join(kenai_3_hash) %>%
  group_by(across(c(-seq_number,-nReads))) %>%
  dplyr::summarise(nReads=sum(nReads)) %>%
  ungroup() %>%
  dplyr::select(-unique_sample, -barcode_1_col, -Miseq_run) %>%
  pivot_wider(names_from=sample, values_from = nReads, values_fill = 0) %>%
  pivot_longer(!sum.taxonomy, names_to = "sample", values_to= "nReads") %>%
 left_join(metadata_kenai_3, by=c("sample"="New_name"))-> kenai_3_asv_pre
```

```
## Joining with `by = join_by(seq_number, sum.taxonomy)`
## `summarise()` has regrouped the output.
```

```
seqrun1_asv_nested <- readRDS(file=here("data","eDNA_data","data","Anacapa_processed_sequence_data","tw

seqrun1_hash <- readRDS(file=here("data","eDNA_data","data","Anacapa_processed_sequence_data","twentymi

seqrun1_asv_nested$Step.4.low.read.depth[[1]] %>%
      left_join(seqrun1_hash) %>%
  group_by(across(c(-seq_number,-nReads))) %>%
  dplyr::summarise(nReads=sum(nReads)) %>%
```

```
  ungroup() %>%
  dplyr::select(-unique_sample, -barcode_1_col, -Miseq_run) %>%
  pivot_wider(names_from=sample, values_from = nReads, values_fill = 0) %>%
  pivot_longer(!sum.taxonomy, names_to = "sample", values_to= "nReads") %>%
 left_join(metadata, by=c("sample"="New_name")) -> seqrun1_asv_pre
```

```
## Joining with `by = join_by(seq_number, sum.taxonomy)`
## `summarise()` has regrouped the output.
```

```
seqrun2_asv_nested <- readRDS(file=here("data","eDNA_data","data","Anacapa_processed_sequence_data","tw

seqrun2_hash <- readRDS(file=here("data","eDNA_data","data","Anacapa_processed_sequence_data","twentymil

seqrun2_asv_nested$Step.4.low.read.depth[[1]] %>%
    left_join(seqrun2_hash) %>%
  group_by(across(c(-seq_number,-nReads))) %>%
  dplyr::summarise(nReads=sum(nReads)) %>%
  ungroup() %>%
  dplyr::select(-unique_sample, -barcode_1_col, -Miseq_run) %>%
  pivot_wider(names_from=sample, values_from = nReads, values_fill = 0) %>%
  pivot_longer(!sum.taxonomy, names_to = "sample", values_to= "nReads") %>%
 left_join(metadata2, by=c("sample"="New_name"))-> seqrun2_asv_pre
```

```
## Joining with `by = join_by(seq_number, sum.taxonomy)`
## `summarise()` has regrouped the output.
```

```
bind_rows(seqrun2_asv_pre, seqrun1_asv_pre, kenai_3_asv_pre,kenai_2_asv_pre,kenai_1_asv_pre) %>%
  filter(., sum.taxonomy !="") %>%
  filter(., !str_detect(sample,"Beluga_Diet")) %>%
  filter(., !str_detect(sample,"Coastal_Even")) %>%
  filter(., !str_detect(sample,"SCSP_")) -> pre_c

pre_c$nReads %>%  min()
```

```
## [1] 0
```

```
# Calculate the total number of reads per sample
pre_c %>%
  group_by(Seq_number,sample) %>%
  dplyr::summarise(total_reads = sum(nReads)) -> tot_reads
```

```
## `summarise()` has regrouped the output.
## i Summaries were computed grouped by Seq_number and sample.
## i Output is grouped by Seq_number.
## i Use `summarise(.groups = "drop_last")` to silence this message.
## i Use `summarise(.by = c(Seq_number, sample))` for per-operation grouping
##   (`?dplyr::dplyr_by`) instead.
```

```r
# Join the total reads with the combined data and calculate the proportion of reads per sample
pre_c %>%
  left_join(tot_reads) %>%
  filter(total_reads > 0) %>% # Filter out samples with zero total reads
  mutate(prop_reads = nReads / total_reads) -> pre_c
```

```
## Joining with `by = join_by(sample, Seq_number)`
```

```r
  pre_c %>%
    filter(., nReads > 0) %>%
  group_by(sample, Seq_number,New_name_no_tech,LabID) %>%
  dplyr::summarise(max_prop_per_sample = max(prop_reads), mean_prop_per_sample = mean(prop_reads),speci
  mutate(., expected_mean_perc_based_on_richness = 1/species_richness,
         expected_richness_based_on_mean_prop = 1/mean_prop_per_sample)-> max_prop_per_sample
```

```
## `summarise()` has regrouped the output.
## i Summaries were computed grouped by sample, Seq_number, New_name_no_tech, and
##   LabID.
## i Output is grouped by sample, Seq_number, and New_name_no_tech.
## i Use `summarise(.groups = "drop_last")` to silence this message.
## i Use `summarise(.by = c(sample, Seq_number, New_name_no_tech, LabID))` for
##   per-operation grouping (`?dplyr::dplyr_by`) instead.
```

```r
library(ggpmisc)
```

```
## Loading required package: ggpp
```

```
## Registered S3 methods overwritten by 'ggpp':
##   method                  from
##   heightDetails.titleGrob ggplot2
##   widthDetails.titleGrob  ggplot2
```

```
##
## Attaching package: 'ggpp'
```

```
## The following objects are masked from 'package:ggpubr':
##
##     as_npc, as_npcx, as_npcy
```

```
## The following object is masked from 'package:ggplot2':
##
##     annotate
```

```r
qPCR_4_species %>%
  mutate(., `Target Name`=if_else(`Target Name`=="cetacea","Beluga",`Target Name`)) %>%
  filter(., `Target Name`!="Beluga") %>%
  mutate(., mean_dnacopy_L=replace_na(mean_dnacopy_L,0)) %>%
  group_by(LabID,`Target Name`) %>%
  dplyr::summarise(mean_mean_dnacopy_L=mean(mean_dnacopy_L)) -> q_mean
```

```
## `summarise()` has regrouped the output.
## i Summaries were computed grouped by LabID and Target Name.
## i Output is grouped by LabID.
## i Use `summarise(.groups = "drop_last")` to silence this message.
## i Use `summarise(.by = c(LabID, Target Name))` for per-operation grouping
##   (`?dplyr::dplyr_by`) instead.
```

```
pre_c %>%
   mutate(., `Target Name`=case_when(str_detect(sum.taxonomy,"Delphinapterus leucas")~ "Beluga" ,
                                     str_detect(sum.taxonomy,"Thaleichthys pacificus") ~ "Eulachon",
                                     str_detect(sum.taxonomy,"Oncorhynchus tshawytscha")~ "Chinook",
                                     str_detect(sum.taxonomy,"Oncorhynchus kisutch") ~ "Coho")) %>%
  filter(., `Target Name`!="Beluga") %>%
   group_by(New_name_no_tech,LabID,`Target Name`, sample, Tech_rep) %>%
  dplyr::summarise(nReads = sum(nReads), prop_reads=max(prop_reads)) %>%
  filter(., !is.na(`Target Name`)) %>%
  ungroup() %>%
    group_by(New_name_no_tech,LabID,`Target Name`) %>%
    dplyr::summarise(prop_reads=mean(prop_reads), mean_reads =mean(nReads),max_reads=max(nReads), min_r
                     max_nReads=max(nReads),
                     non_na_tech_count = n_distinct(sample),
                     non_na_tech_count2 = n_distinct(Tech_rep),
          Drop_outs = length(nReads[nReads==0]),
    drop_prop=Drop_outs/non_na_tech_count/non_na_tech_count2) %>%
  mutate(., drop_prop=if_else(drop_prop>1,1,drop_prop)) -> dropout_prop_all_pre
```

```
## `summarise()` has regrouped the output.
## i Summaries were computed grouped by New_name_no_tech, LabID, Target Name,
##   sample, and Tech_rep.
## i Output is grouped by New_name_no_tech, LabID, Target Name, and sample.
## i Use `summarise(.groups = "drop_last")` to silence this message.
## i Use `summarise(.by = c(New_name_no_tech, LabID, Target Name, sample,
##   Tech_rep))` for per-operation grouping (`?dplyr::dplyr_by`) instead.
```

```
## `summarise()` has regrouped the output.
## i Summaries were computed grouped by New_name_no_tech, LabID, and Target Name.
## i Output is grouped by New_name_no_tech and LabID.
## i Use `summarise(.groups = "drop_last")` to silence this message.
## i Use `summarise(.by = c(New_name_no_tech, LabID, Target Name))` for
##   per-operation grouping (`?dplyr::dplyr_by`) instead.
```

```
dropout_prop_all_pre %>%
  filter(., !is.na(max_nReads))-> plotting_drop_outs_all

plotting_drop_outs_all %>%
    left_join(q_mean) %>%
  filter(., mean_mean_dnacopy_L > 1) %>%
  filter(., drop_prop ==1)  -> concerning_drop_outs
```

```
## Joining with `by = join_by(LabID, `Target Name`)`
```

```
combo_4_species %>%
  filter(., mean_dnacopy_L > 1 & index_scores == 0) -> high_qPCR_no_edna_detection

high_qPCR_no_edna_detection %>%
  dplyr::select(Sample, `Target Name`) %>%
  distinct() -> sample_w_problems_species

high_qPCR_no_edna_detection$Sample %>%  unique() -> sample_w_problems

combined_data_long_reads_c %>%
  dplyr::select(Sample, Seq_number,New_name_no_tech) %>% distinct() %>%
  left_join(sample_w_problems_species) %>%
  filter(., !is.na(`Target Name`))-> sample_w_problems_tib
```

```
## Joining with ‘by = join_by(Sample)‘

## Warning in left_join(., sample_w_problems_species): Detected an unexpected many-to-many relationship
## i Row 138 of ‘x‘ matches multiple rows in ‘y‘.
## i Row 6 of ‘y‘ matches multiple rows in ‘x‘.
## i If a many-to-many relationship is expected, set ‘relationship =
##   "many-to-many"‘ to silence this warning.
```

```
combined_data_long_reads_c %>%
  dplyr::select(Sample, Seq_number,New_name_no_tech) %>% distinct() %>%
  left_join(concerning_drop_outs) %>%
  filter(., !is.na(`Target Name`))-> sample_w_concerns_tib
```

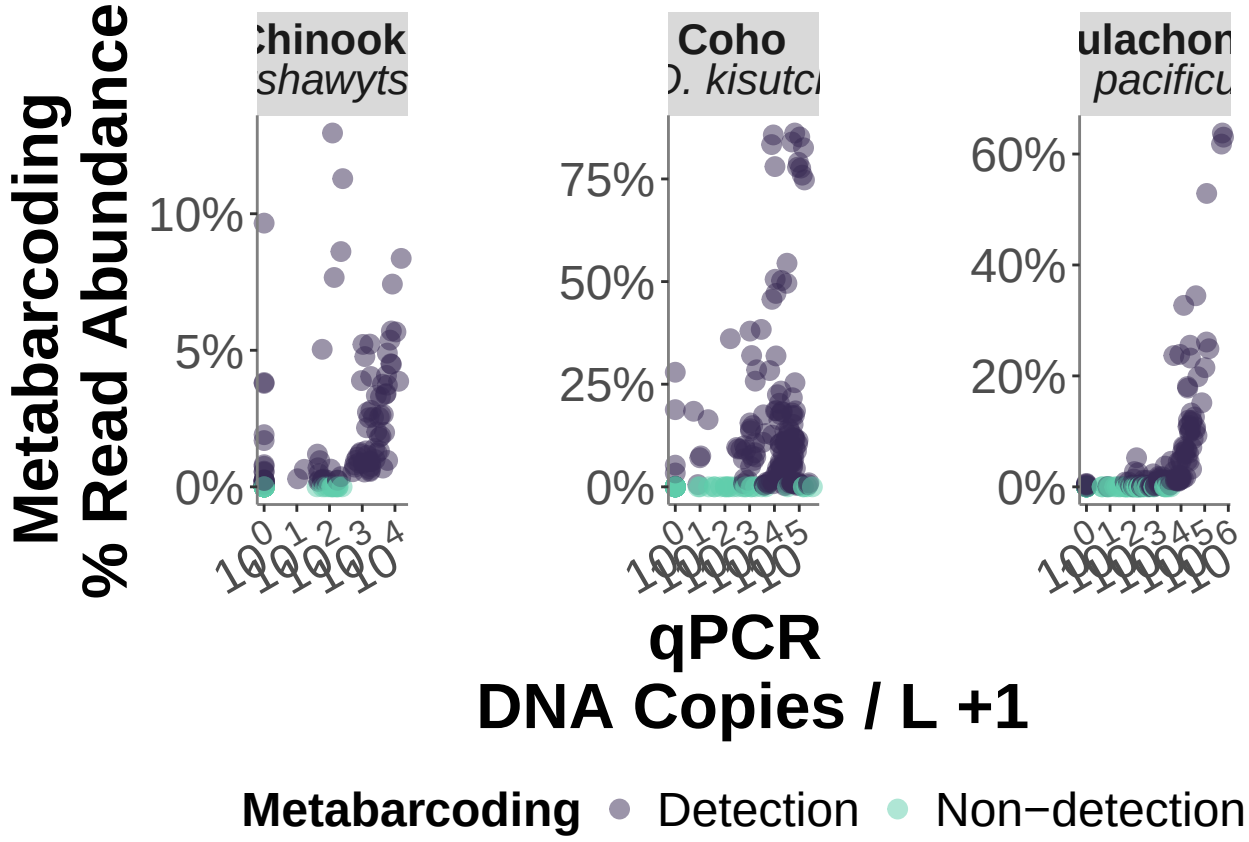
```
## Joining with ‘by = join_by(New_name_no_tech)‘

## Warning in left_join(., concerning_drop_outs): Detected an unexpected many-to-many relationship betwe
## i Row 139 of ‘x‘ matches multiple rows in ‘y‘.
## i Row 18 of ‘y‘ matches multiple rows in ‘x‘.
## i If a many-to-many relationship is expected, set ‘relationship =
##   "many-to-many"‘ to silence this warning.
```

```
dropout_prop_all_pre %>%
  left_join(q_mean) %>%
     filter(., !is.na(`Target Name`)) %>%
    mutate(., drop_out= if_else(New_name_no_tech %in% sample_w_problems_tib$New_name_no_tech,New_name_n
  #filter(., drop_prop >0) %>%
arrange(mean_mean_dnacopy_L) %>%
filter(., drop_prop %in% c(0,1)) %>%
mutate(., drop_bin=if_else(drop_prop>0,"Non-detection","Detection")) %>%
mutate(., `Target Name`=case_when(`Target Name`=="Beluga"~ "Beluga \n *D. leucas*" ,
                                  `Target Name`=="Eulachon" ~ "**Eulachon** <br> *T. pacificus*",
                                  `Target Name`=="Chinook"~ "**Chinook** <br> *O. tshawytscha*",
                                  `Target Name`=="Coho" ~ "**Coho** <br> *O. kisutch*")) %>%
   ggplot(., aes(x=mean_mean_dnacopy_L+1, colour=drop_bin, y=prop_reads))+
      geom_point(alpha =0.5, size=3)  +
  ylab("Metabarcoding \n % Read Abundance") +xlab("qPCR \n DNA Copies / L +1")   +
theme(axis.text.x = element_text(angle=30, hjust = 1), panel.grid.major = element_blank(), panel.grid.m
```

```
panel.background = element_blank(), axis.text = element_text(size=18), axis.title = element_text(size=24
  #geom_text(aes(label=drop_out)) +
  facet_wrap(`Target Name`~., scales="free") +
  scale_y_continuous(labels = scales::label_percent()) +
  scale_color_viridis_d(option="mako", begin=0.2, end =0.8) +
  labs(color = "Metabarcoding")+  coord_cartesian(clip = "off") +
  scale_x_log10(
    breaks = c(1, 10, 100, 1000, 10000, 100000, 1000000),
    labels = trans_format("log10", math_format(10^.x))
  ) + theme(strip.text = element_markdown())
```

```
## Joining with `by = join_by(LabID, `Target Name`)`
```



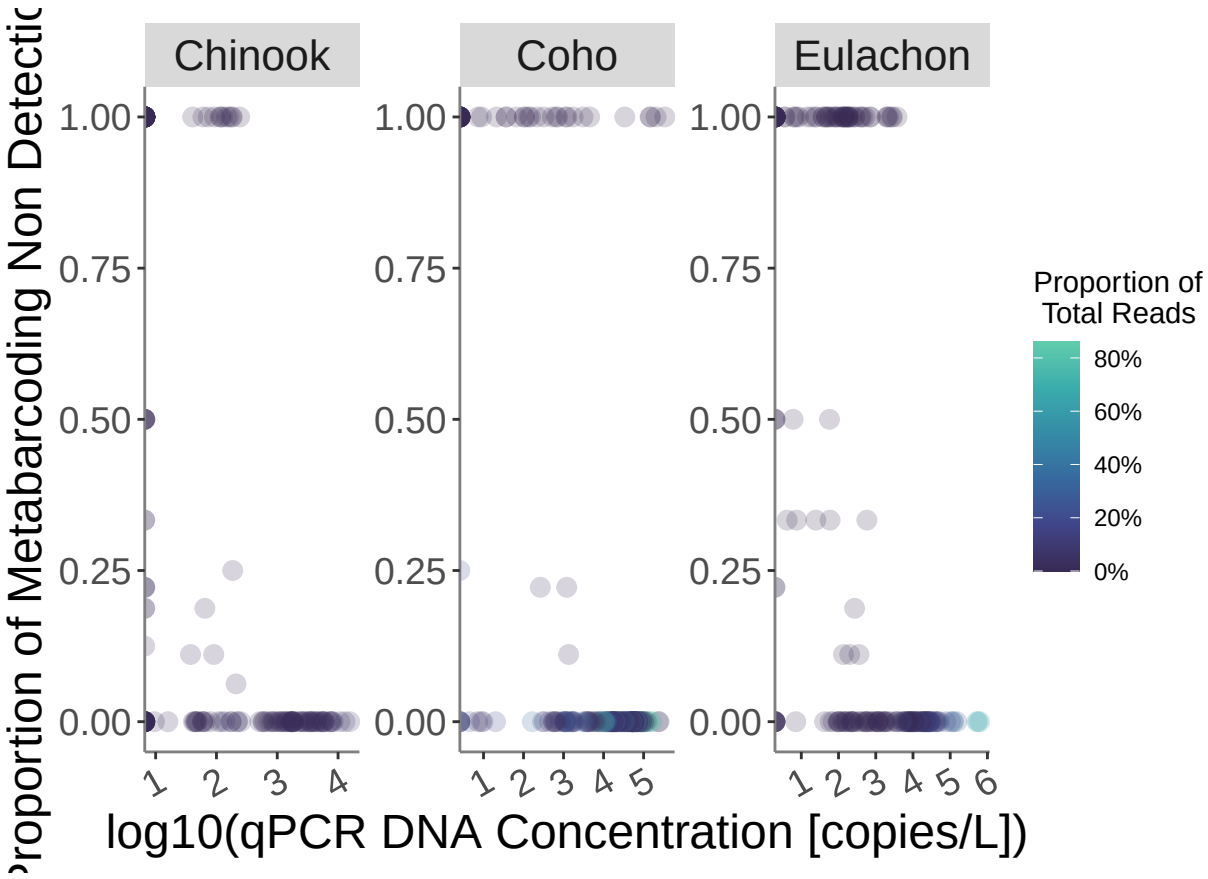

```
ggsave(file = here::here("analysis", "figures", "Figure_7_swamping_v2.png"),
 width = 12,
 height = 8
 )
```

**Figure S12. Species Swamping**

```
dropout_prop_all_pre %>%
  left_join(q_mean) %>%
```

```
    filter(., !is.na(`Target Name`)) %>%
    mutate(., drop_out= if_else(New_name_no_tech %in% sample_w_problems_tib$New_name_no_tech,New_name_n
  #filter(., drop_prop >0) %>%
arrange(desc(mean_mean_dnacopy_L)) %>%
   ggplot(., aes(x=log10(mean_mean_dnacopy_L), y=drop_prop, colour=prop_reads))+
     geom_point(alpha =0.2, size=3)  +facet_wrap(`Target Name`~., scale="free") +
  ylab("Proportion of Metabarcoding Non Detections") +xlab("log10(qPCR DNA Concentration [copies/L])")
theme(axis.text.x = element_text(angle=30, hjust = 1), panel.grid.major = element_blank(), panel.grid.mi
panel.background = element_blank(), axis.text = element_text(size=14), axis.title = element_text(size=18
  #geom_text(aes(label=drop_out)) +
  facet_wrap(`Target Name`~., scales="free") +
  scale_color_viridis_c(option="mako", begin=0.2, end =0.8, labels = scales::percent) +
  labs(color = "Proportion of \n Total Reads")
```

```
## Joining with `by = join_by(LabID, `Target Name`)`
```




```
ggsave(file = here::here("analysis", "figures", "Figure_S12_swamping.png"),
 width = 8,
 height = 6
 )
```

```
dropout_prop_all_pre %>%
  left_join(q_mean) %>%
    filter(., !is.na(`Target Name`)) %>%
```

```
    mutate(., drop_out= if_else(New_name_no_tech %in% sample_w_problems_tib$New_name_no_tech,New_name_n
  group_by(`Target Name`) %>%
  filter(., Drop_outs >0) %>%
  filter(., mean_mean_dnacopy_L >0) %>%
  dplyr::summarise(n_distinct(New_name_no_tech), max(max_reads), mean(mean_reads), max(mean_mean_dnacopy
```

```
## Joining with `by = join_by(LabID, `Target Name`)`
```

```
## # A tibble: 3 x 6
##   `Target Name` n_distinct(New_name_no_tec~1 `max(max_reads)` `mean(mean_reads)`
##   <chr>                                <int>            <dbl>              <dbl>
## 1 Chinook                                 17             1270             105.
## 2 Coho                                    28             2056              41.4
## 3 Eulachon                                60              324               4.33
## # i abbreviated name: 1: `n_distinct(New_name_no_tech)`
## # i 2 more variables: `max(mean_mean_dnacopy_L)` <dbl>,
## #   `mean(mean_mean_dnacopy_L)` <dbl>
```

```
dropout_prop_all_pre %>%
  left_join(q_mean) %>%
     filter(., !is.na(`Target Name`)) %>%
    mutate(., drop_out= if_else(New_name_no_tech %in% sample_w_problems_tib$New_name_no_tech,New_name_n
  group_by(`Target Name`) %>%
  filter(., Drop_outs >0) %>%
  filter(., `Target Name`=="Coho") %>%
  arrange(desc(mean_mean_dnacopy_L))
```

```
## Joining with `by = join_by(LabID, `Target Name`)`
```

```
## # A tibble: 61 x 14
## # Groups:   Target Name [1]
##    New_name_no_tech     LabID      `Target Name` prop_reads mean_reads max_reads
##    <chr>                <chr>      <chr>              <dbl>      <dbl>     <dbl>
##  1 Bridge_Surface_Jul_3 52386_138~ Coho             0               0         0
##  2 Bridge_5m_Jul_2      52386_140  Coho             0               0         0
##  3 Bridge_5m_Jul_1      52386_139  Coho             0               0         0
##  4 Bridge_Surface_Jul_2 52386_137~ Coho             0               0         0
##  5 Bridge_Surface_Jul_1 52386_136~ Coho             0               0         0
##  6 Dock_Surface_Jun_2   52386_116  Coho             0               0         0
##  7 Dock_5m_Jun_1        52386_118  Coho             0               0         0
##  8 Bridge_5m_May_2      52386_112  Coho             0               0         0
##  9 Dock_Surface_May_3   52386_103  Coho             0.00672       192.      299
## 10 Bridge_5m_Nov.21_3   52386_198  Coho             0               0         0
## # i 51 more rows
## # i 8 more variables: min_reads <dbl>, max_nReads <dbl>,
## #   non_na_tech_count <int>, non_na_tech_count2 <int>, Drop_outs <int>,
## #   drop_prop <dbl>, mean_mean_dnacopy_L <dbl>, drop_out <chr>
```

```
combined_data_long_reads_c %>%
  filter(., River =="Kenai" & str_detect(New_name_no_tech, "Jun|Jul")) %>%
  filter(., nReads >0) %>%
```

```
    arrange(desc(prop_reads)) %>%
  filter(., Species=="Oncorhynchus nerka") %>%
  dplyr::summarise(max(prop_reads), min(prop_reads), mean(prop_reads))
```

```
## # A tibble: 1 x 3
##   `max(prop_reads)` `min(prop_reads)` `mean(prop_reads)`
##               <dbl>             <dbl>              <dbl>
## 1             0.894             0.166              0.659
```

```
combined_data_long_reads_c %>%
  filter(., River =="Kenai" & str_detect(New_name_no_tech, "Jun|Jul")) %>%
  filter(., nReads >0) %>%
  group_by(New_name_no_tech) %>%
  dplyr::summarise(n_distinct(Species)) %>%
  ungroup() %>%
  dplyr::summarise(mean(`n_distinct(Species)`))
```

```
## # A tibble: 1 x 1
##   `mean(\`n_distinct(Species)\`)`
##                             <dbl>
## 1                            10.5
```

```
combined_data_long_reads_c %>%
  filter(., River =="Kenai" & !str_detect(New_name_no_tech, "Jun|Jul")) %>%
  filter(., nReads >0) %>%
  group_by(New_name_no_tech) %>%
  dplyr::summarise(n_distinct(Species)) %>%
  ungroup() %>%
  dplyr::summarise(mean(`n_distinct(Species)`))
```

```
## # A tibble: 1 x 1
##   `mean(\`n_distinct(Species)\`)`
##                             <dbl>
## 1                            11.2
```

```
sample_w_concerns_tib %>%
    left_join(pre_c) %>%
  mutate(., creek= River) %>%
  filter(., creek %in% c("Twentymile")) %>%
  mutate(., Sample_date =as.Date(Sample_date, format = "%m/%d/%y")) %>%
  mutate(time_chunk = case_when(Sample_date < '2021-08-16' ~"Before",
    Sample_date > "2021-10-21" ~"After",
    TRUE ~"During")) %>%
 mutate(., Units ="copies/L") %>%
  mutate(.,bottle =str_c(Sample_date,"_", creek,"_4","_",Bio_rep ) )-> sample_w_concerns_tib_analysis
```

```
## Joining with `by = join_by(Seq_number, New_name_no_tech, LabID, prop_reads)`
```

```
## Warning in left_join(., pre_c): Detected an unexpected many-to-many relationship between `x` and `y`
## i Row 1 of `x` matches multiple rows in `y`.
## i Row 16616 of `y` matches multiple rows in `x`.
## i If a many-to-many relationship is expected, set `relationship =
##   "many-to-many"` to silence this warning.
```

```
sample_w_problems_tib %>%
  left_join(pre_c) %>%
  mutate(., creek= River) %>%
  filter(., creek %in% c("Twentymile")) %>%
  mutate(., Sample_date =as.Date(Sample_date, format = "%m/%d/%y")) %>%
  mutate(time_chunk = case_when(Sample_date < '2021-08-16' ~"Before",
    Sample_date > "2021-10-21" ~"After",
    TRUE ~"During")) %>%
 mutate(., Units ="copies/L") %>%
  mutate(.,bottle =str_c(Sample_date,"_", creek,"_4","_",Bio_rep ) )-> sample_w_problems_tib_analysis
```

```
## Joining with `by = join_by(Seq_number, New_name_no_tech)`
```

```
## Warning in left_join(., pre_c): Detected an unexpected many-to-many relationship between `x` and `y`
## i Row 1 of `x` matches multiple rows in `y`.
## i Row 16461 of `y` matches multiple rows in `x`.
## i If a many-to-many relationship is expected, set `relationship =
##   "many-to-many"` to silence this warning.
```

```
 pre_c %>%
  mutate(., creek= River) %>%
  filter(., creek %in% c("Twentymile")) %>%
  mutate(., Sample_date =as.Date(Sample_date, format = "%m/%d/%y")) %>%
  mutate(time_chunk = case_when(Sample_date < '2021-08-16' ~"Before",
    Sample_date > "2021-10-21" ~"After",
    TRUE ~"During")) %>%
 mutate(., Units ="copies/L") %>%
  mutate(.,bottle =str_c(Sample_date,"_", creek,"_4","_",Bio_rep ) ) %>%
  mutate(., drop_out_problem= if_else(bottle %in% sample_w_problems_tib_analysis$bottle,"Yes","No")) %>%
   mutate(., drop_out_concerns= if_else(bottle %in% sample_w_concerns_tib_analysis$bottle,"Yes","No")) -

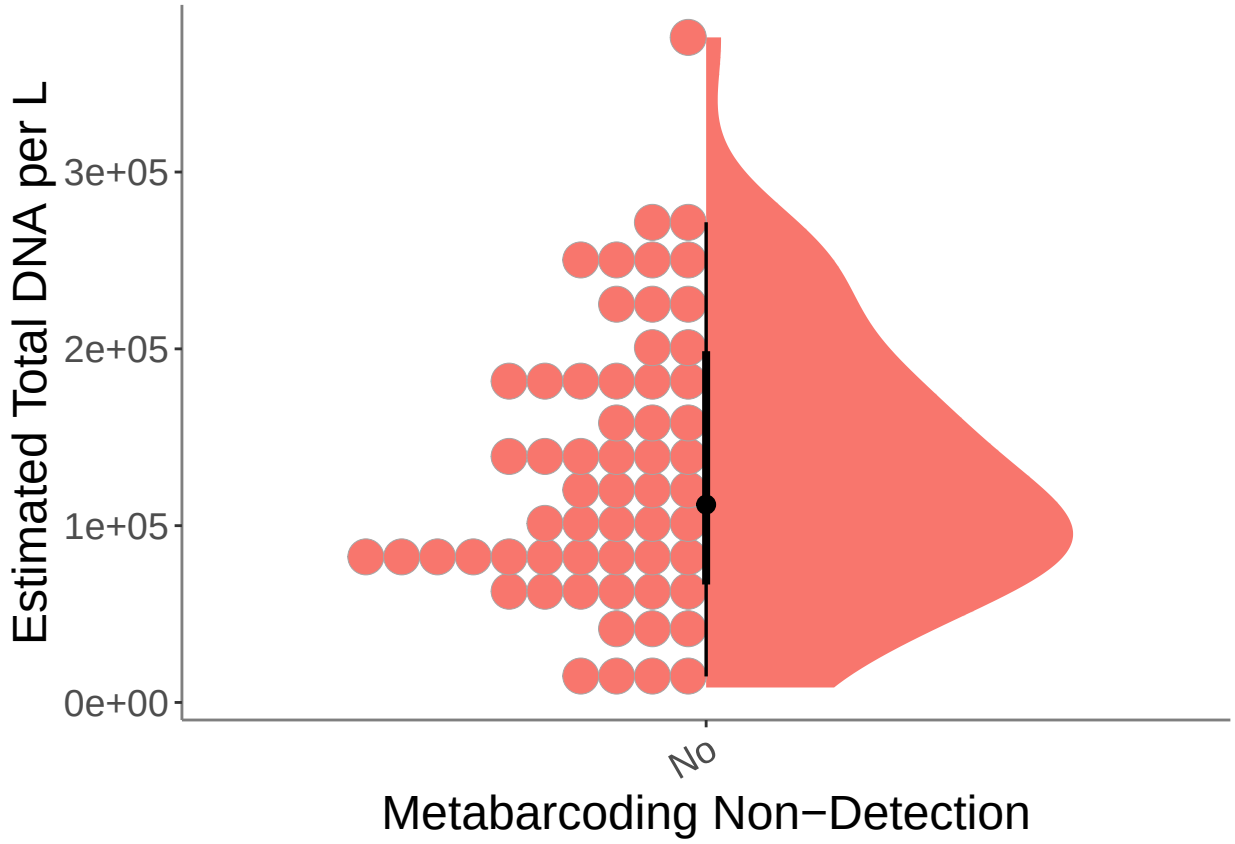

coho_anchor_20m %>%
  left_join(meta_c_analysis) %>%
  group_by(bottle, drop_out_concerns, Beluga) %>%
  dplyr::summarise(est_tot_dna = mean(meantotdna_coho)) %>%
  arrange(est_tot_dna) %>%
  ggplot(., aes(y=est_tot_dna, x=drop_out_concerns, fill= drop_out_concerns)) + stat_slab(aes(thickness
  stat_dotsinterval(side = "bottom", scale = 0.7, slab_linewidth = NA)  + ylab("Estimated Total DNA per
panel.background = element_blank(), axis.text = element_text(size=14), axis.title = element_text(size=18
```

```
## Joining with `by = join_by(bottle, creek, Sample_date)`
```

```
## Warning in left_join(., meta_c_analysis): Detected an unexpected many-to-many relationship between `
## i Row 1 of `x` matches multiple rows in `y`.
## i Row 2611 of `y` matches multiple rows in `x`.
## i If a many-to-many relationship is expected, set `relationship =
##   "many-to-many"` to silence this warning.
```

```
## `summarise()` has regrouped the output.
## i Summaries were computed grouped by bottle, drop_out_concerns, and Beluga.
```

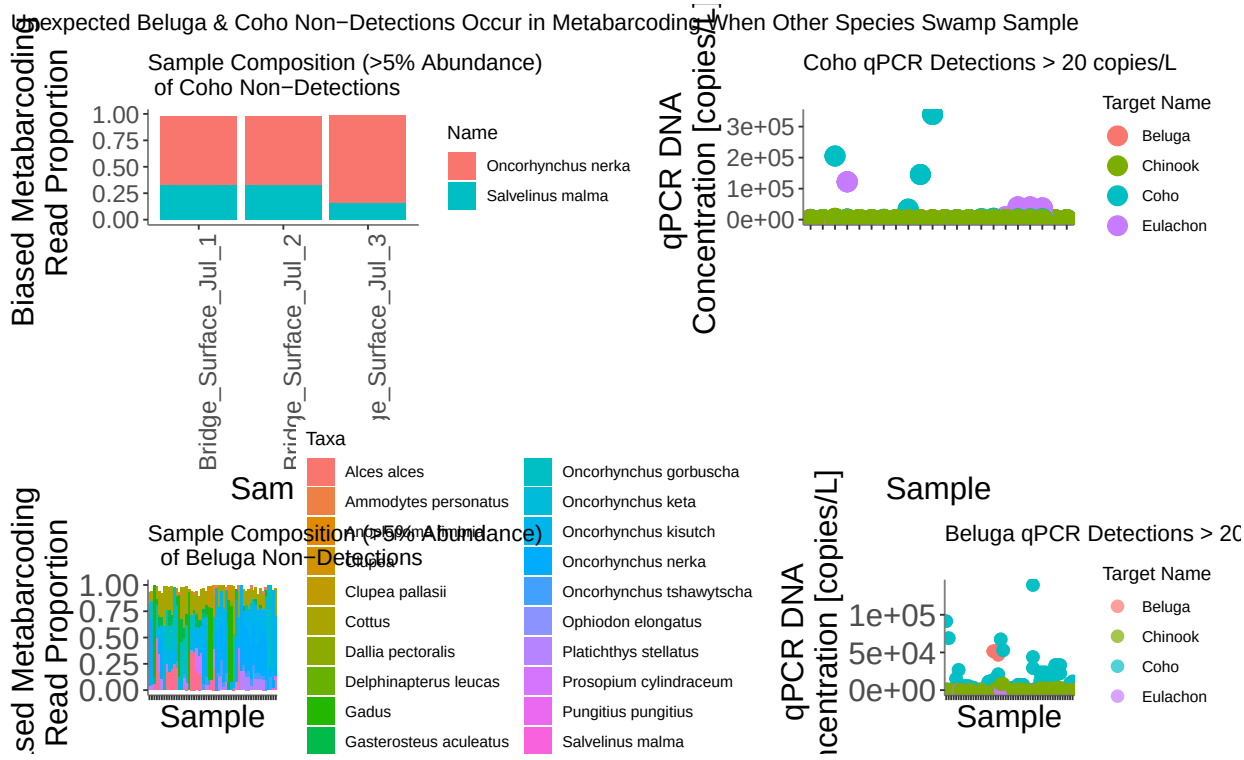
```
## i Output is grouped by bottle and drop_out_concerns.
## i Use `summarise(.groups = "drop_last")` to silence this message.
## i Use `summarise(.by = c(bottle, drop_out_concerns, Beluga))` for per-operation
##   grouping (`?dplyr::dplyr_by`) instead.
```




```
ggsave(file = here::here("analysis", "figures", "Figure_S4_swamping_tot_dna.png"),
 width = 12.5,
 height = 10
 )
```

```
## # A tibble: 17 x 4
##    New_name   `mean(mean_dnacopy_L)` `max(mean_dnacopy_L)` `min(mean_dnacopy_L)`
##    <chr>                       <dbl>                <dbl>                  <dbl>
##  1 Bridge_5m~                  104.                 156.                     0
##  2 Bridge_5m~                  716.                 716.                   716.
##  3 Bridge_5m~               205148.              205148.                205148.
##  4 Bridge_Su~                  148.                 525.                     0
##  5 Bridge_Su~                  323.                 323.                   323.
##  6 Bridge_Su~                 1130.                1130.                  1130.
##  7 Bridge_Su~                    7.34                22.0                    0
##  8 Bridge_Su~                33841.               33841.                 33841.
##  9 Bridge_Su~               145644.              145644.                145644.
## 10 Bridge_Su~               338892.              338892.                338892.
## 11 Bridge_Su~                   21.2                 63.6                    0
## 12 Dock_5m_A~                  106.                 158.                     0
```

```
## 13 Dock_5m_A~              36.8              110.                 0
## 14 Dock_5m_J~            3063.              3063.              3063.
## 15 Dock_Surf~            4496.              4496.              4496.
## 16 Susitna_0~             527.               527.               527.
## 17 Susitna_0~              66.5               99.8                0
```




Here we focus on samples that had greater than 20 copies/L of DNA as measured by qPCR and non-detections via metabarcoding for Coho (top) and Beluga (bottom). On the left we report the raw biased proportion of observed metabarcoding reads and on the right the corresponding qPCR concentrations.

Here we consistently observe non-detection occur when a single species constitutes $> 30\%$ of all reads for Coho and $> 40\%$ for Beluga.

We note that for Coho we had non-detections during the highest observed concentrations from eDNA samples during the height of the Kenai River sockeye run. The two May time points with relatively high Eulachon abundance also had non-detections. We note that Eulachon had the lowest observed amplification efficiency of any species in our mock community and thus even a 30% abundance represents an enormous concentration of Eulachon DNA molecules in the sample likely leading to competitive interactions that resulted in Coho non-detections (See Figure S1).

The results align to the results of Gold et al. 2023 (Gold et al. 2023) aside from the clear instances of swamping.

```
plotting_drop_outs_all %>%
    left_join(q_mean) %>%
  mutate(., Detection = case_when(mean_mean_dnacopy_L ==0 & mean_reads ==0 ~"Both Absent",
                                  mean_mean_dnacopy_L ==0 & mean_reads >0 ~"Metabarcoding Only",
                                  mean_mean_dnacopy_L >0 & mean_reads ==0 ~"qPCR Only",
                                  mean_mean_dnacopy_L >0 & mean_reads >0 ~"Both Detected",
)) %>%
  filter(., !is.na(Detection)) %>%
```

```
  group_by(`Target Name`) %>%
  dplyr::count(Detection) %>%
  group_by(`Target Name`) %>%
  mutate(., tot=sum(n))
```

```
## Joining with `by = join_by(LabID, `Target Name`)`
```

```
## # A tibble: 12 x 4
## # Groups:   Target Name [3]
##    `Target Name` Detection             n   tot
##    <chr>         <chr>             <int> <int>
##  1 Chinook       Both Absent          87   210
##  2 Chinook       Both Detected        83   210
##  3 Chinook       Metabarcoding Only   28   210
##  4 Chinook       qPCR Only            12   210
##  5 Coho          Both Absent          32   210
##  6 Coho          Both Detected       147   210
##  7 Coho          Metabarcoding Only    6   210
##  8 Coho          qPCR Only            25   210
##  9 Eulachon      Both Absent          34   210
## 10 Eulachon      Both Detected       110   210
## 11 Eulachon      Metabarcoding Only   10   210
## 12 Eulachon      qPCR Only            56   210
```

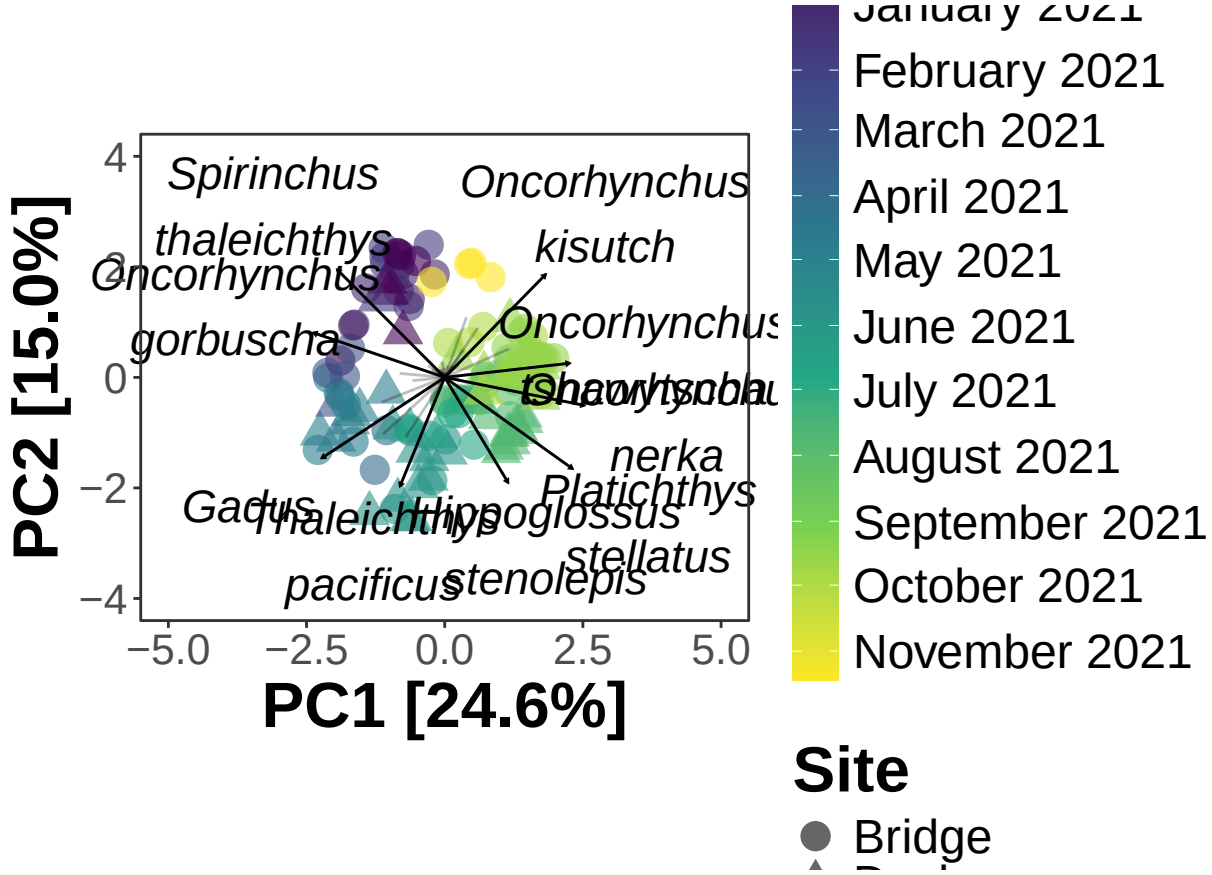
Here we show that qPCR and metabarcoding were largely concordant across all samples for the four focal species. However, qPCR more consistently detected species than metabarcoding.

**Figure 8. PCA of Temporal and Spatial Variation**

```
tax_table(physeq_obj_kenai_reads)[, "ID_mifish"] <- str_replace_all(tax_table(physeq_obj_kenai_reads)[,
sample_data(physeq_obj_kenai_reads)$Sample_date2 <- as.Date(sample_data(physeq_obj_kenai_reads)$Sample_

physeq_obj_kenai_reads %>%
  tax_transform("clr", rank="ID_mifish") %>%
  ord_calc(method = "PCA") %>%
  ord_plot(color = "Sample_date2",
           shape ="Site",
           size = 5,
           alpha=0.6,
           plot_taxa = 1:9,
           auto_caption=NA,
           tax_vec_length = 1,
           tax_lab_length = 1.6,
           tax_lab_style = tax_lab_style(size = 6,  type = "text", fontface = "italic"))  +
  theme_bw()+
  theme(plot.title = element_text(hjust = 0.5,size = 20, face = "bold"),
        plot.subtitle = element_text(hjust = 0.5,size = 18, face = "bold"),
        panel.grid.major = element_blank(),
        panel.grid.minor = element_blank(),
        panel.background = element_blank(),
        axis.text=element_text(size=16),
```

```
      axis.title=element_text(size=24,face="bold"),
      legend.title = element_text( size=24, face="bold"),
      legend.text = element_text(size=18),
      ) +
coord_fixed(ratio = 1, clip = "off", xlim = c(-5, 5), ylim = c(-4, 4)) +
labs(shape = "Site", color = "Date Sampled") +
scale_color_viridis_c(
  trans = "date",
  breaks = breaks_width("1 month"),
  labels = label_date("%B %Y"), # Formats legend as "November 2026",
  guide = guide_colorbar(barheight = unit(10, "cm"),
    reverse = TRUE)
)
```

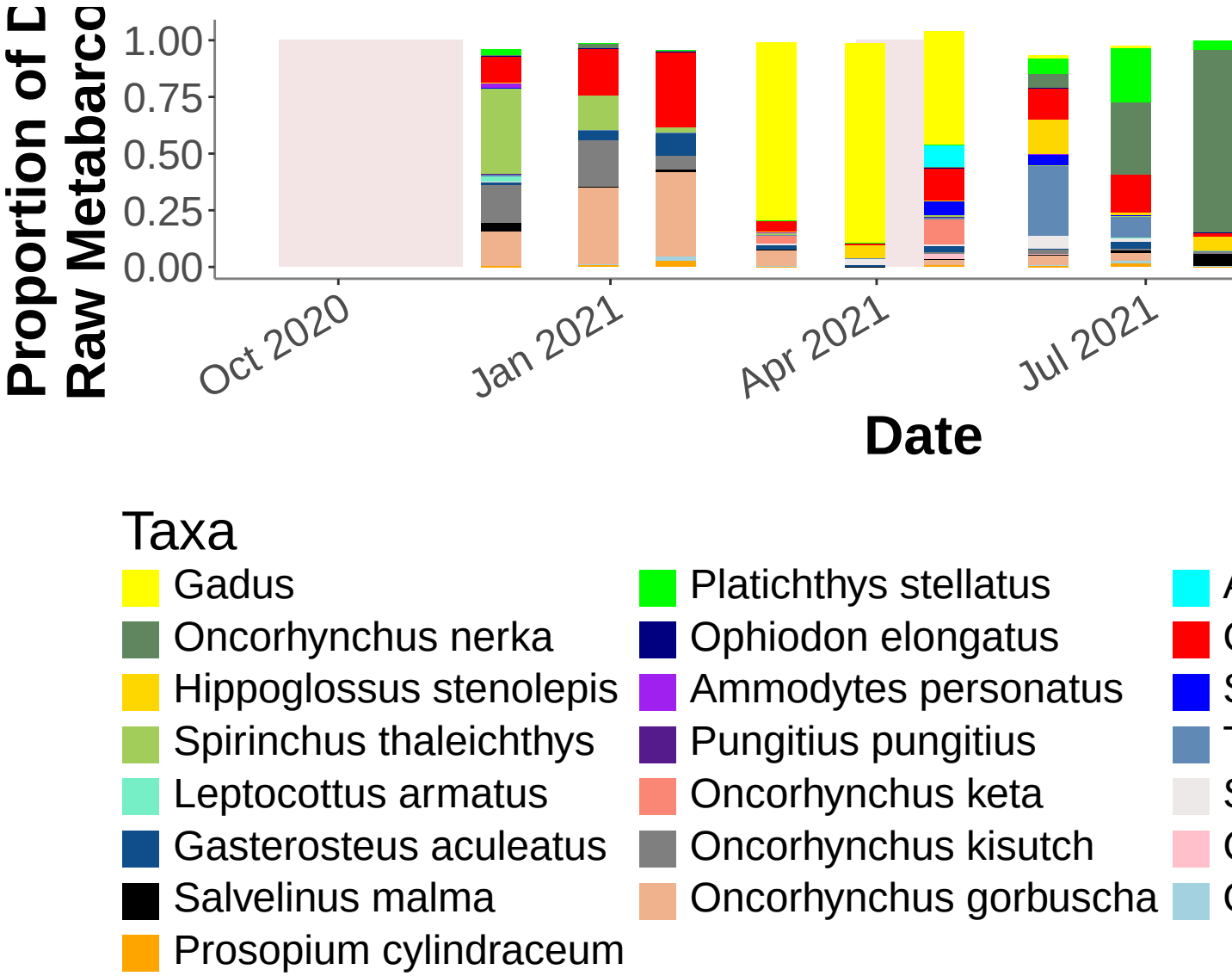



```
ggsave(file = here::here("analysis", "figures", "Figure_8_PCA_aichison.png"),
 width = 12.5,
 height = 10
 )
```

We observe clustering associated with date sampled and to a lesser extent site sampled.

### Kenai Metabarcoding

**Figure S6.7 eDNA Raw Metabarcoding Results**




## Conclusions

Together, these results provide important baseline information needed to inform effective eDNA sampling efforts needed to resolve Cook Inlet Beluga habitat use. Specifically, these findings suggest that weekly sampling efforts at a single site along multiple important rivers would provide sufficient resolution to identify spatial and temporal river foraging habitat utilization of Cook Inlet Beluga whales. These results demonstrate that well calibrated and designed eDNA approaches are an effective tool for capturing changes of Beluga prey assemblages across both space and time. Here we recommend future efforts employ

1. a multiplexed qPCR assay that targets Beluga, King, Coho, Sockeye, Pink, Chum, and Eulachon
2. MiFish 12S metabarcoding as employed here
3. more robust mock communities targeting the 25 fish species observed in Cook Inlet

4. larger pore size sampling (5 µm) to ease sample filtration in high sediment load environments.

Ultimately, our results demonstrate that targeted deployments of eDNA across Cook Inlet habits can reveal critical ecological insights into fish ecology and endangered Cook Inlet Beluga habitat use.

# References


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